✦ Memory Management: Paging • Segmentation • Virtual Memory

TABLE OF CONTENTS

#	Topic / Section	Levels Covered
1	Operating System – Basics & Functions	Easy • Medium • Hard
2	Process Management – FCFS Scheduling	Easy • Medium • Hard
3	Process Management – Round Robin Scheduling	Easy • Medium • Hard
4	Process Management – Priority-Based Scheduling	Easy • Medium • Hard
5	Memory Management – Paging	Easy • Medium • Hard
6	Memory Management – Segmentation	Easy • Medium • Hard
7	Memory Management – Virtual Memory	Easy • Medium • Hard
8	I/O Management – Concept of I/O Port	Easy • Medium • Hard
9	File Management – FAT & File Handling Functions	Easy • Medium • Hard
10	Interrupts – Software & Hardware	Easy • Medium • Hard
11	I/O and Memory-Based Addressing	Easy • Medium • Hard
12	DMA – Direct Memory Access	Easy • Medium • Hard
13	Numerical & Calculation MCQs	Easy • Medium • Hard
14	Assertion-Reason & Tricky MCQs	Easy • Medium • Hard
15	Fill-in-the-Blank & One-Liner MCQs	Easy • Medium • Hard
16	OS Basics – System Calls & Kernel Types	Easy • Medium • Hard
17	Process Management – SJF & SRTF	Easy • Medium • Hard
18	Memory Management – Additional Concepts	Easy • Medium • Hard
19	File Management – Extended Concepts	Easy • Medium • Hard
20	Interrupts – Advanced Concepts	Easy • Medium • Hard
21	DMA – Advanced & Application Concepts	Easy • Medium • Hard
22	Mixed Revision – All Topics	Easy • Medium • Hard

OPERATING SYSTEM – BASICS & FUNCTIONS

■ EASY LEVEL QUESTIONS

1. What is an Operating System?

A) A collection of application programs
B) Software that manages hardware and software resources
C) A hardware component of a computer
D) A type of programming language

Correct Answer: B)

Explanation: An OS is system software that manages hardware, software resources, and provides common services. It acts as an intermediary between users and hardware.

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2. Which of the following is NOT a function of an Operating System?

A) Process Management
B) Memory Management
C) Compiling source code
D) File Management

Correct Answer: C)

Explanation: Compiling source code is a function of a compiler, not the OS. OS functions include process, memory, file, I/O management, and security.

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3. The OS acts as an interface between:

A) Hardware and compiler
B) User and hardware
C) Compiler and assembler
D) Application software and compiler

Correct Answer: B)

Explanation: The OS acts as an interface between user and computer hardware, translating user commands into hardware instructions.

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4. Which component of OS directly interacts with hardware?

A) Shell
B) Compiler
C) Kernel
D) GUI

Correct Answer: C)

Explanation: The Kernel is the core of the OS that directly interacts with hardware — managing CPU, memory, and device drivers.

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5. The command interpreter of an OS is called:

A) Kernel
B) Shell
C) Compiler
D) Linker

Correct Answer: B)

Explanation: The Shell is the command interpreter — it reads user commands and passes them to the kernel. Examples: Bash, PowerShell, CMD.

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6. Which type of OS allows multiple users to use the computer simultaneously?

A) Batch OS
B) Real-time OS
C) Multi-user OS
D) Embedded OS

Correct Answer: C)

Explanation: Multi-user OS allows multiple users to access resources simultaneously. Examples: UNIX, Linux.

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7. A Batch OS processes jobs:

A) With user interaction for each job
B) In a batch without user interaction
C) One byte at a time
D) In real-time

Correct Answer: B)

Explanation: Batch OS collects similar jobs into batches and processes them sequentially without user interaction. Used in early mainframes.

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8. Which OS type guarantees response within a specific time constraint?

A) Batch OS
B) Time-sharing OS
C) Real-time OS
D) Distributed OS

Correct Answer: C)

Explanation: Real-Time OS (RTOS) guarantees task completion within specified time constraints. Used in aircraft control, medical devices, and industrial automation.

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9. The nucleus or core of an OS is called:

A) Shell
B) Kernel
C) API
D) Bootloader

Correct Answer: B)

Explanation: The Kernel is the nucleus/core of the OS — the first program loaded into memory with complete control over the system.

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10. Booting refers to:

A) Shutting down the computer
B) Loading the OS into memory at startup
C) Compiling the OS code
D) Defragmenting the disk

Correct Answer: B)

Explanation: Booting (bootstrapping) is the process of loading the OS kernel into RAM when the computer is powered on. BIOS/UEFI starts the bootloader which loads the OS.

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11. Which type of OS is used in ATMs and washing machines?

A) Batch OS
B) Real-time OS
C) Distributed OS
D) Network OS

Correct Answer: B)

Explanation: ATMs, washing machines, and medical equipment use Real-time OS (RTOS) because they require guaranteed response within strict time constraints.

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12. GUI stands for:

A) General User Interface
B) Graphical User Interface
C) Generic Utility Interface
D) Global Unix Interface

Correct Answer: B)

Explanation: GUI = Graphical User Interface. Provides visual icons, menus, and windows for interaction, as opposed to CLI (Command Line Interface).

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13. SPOOL stands for:

A) Simultaneous Peripheral Operations On-Line
B) Serial Processing of Output Linking
C) System Program Output Loading
D) Simultaneous Program Output Logging

Correct Answer: A)

Explanation: SPOOL = Simultaneous Peripheral Operations On-Line. Buffers slow-device data (like printers) on disk, allowing CPU to continue without waiting.

■ MEDIUM LEVEL QUESTIONS

14. System calls provide an interface between:

A) User and Shell
B) Kernel and Shell
C) Process and OS
D) Hardware and Shell

Correct Answer: C)

Explanation: System calls provide the interface between a running process and the OS kernel — allowing programs to request OS services.

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15. PCB (Process Control Block) contains:

A) File data
B) Process state, PID, registers, memory limits
C) Application source code
D) Disk sectors

Correct Answer: B)

Explanation: PCB stores: PID, process state, program counter, CPU registers, memory limits, open file list, I/O status, and scheduling information.

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16. The state where a process is ready to run but waiting for CPU is called:

A) Running state
B) Waiting state
C) Ready state
D) Terminated state

Correct Answer: C)

Explanation: Ready state: process is in memory with all resources allocated, waiting only for CPU. Scheduler moves it to Running state.

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17. Context switching saves and restores:

A) Disk data
B) Process state (registers, PC, stack pointer)
C) File handles only
D) Network connections

Correct Answer: B)

Explanation: Context switching saves the current process's PCB (registers, PC, stack pointer) and restores the next process's state. It is pure OS overhead.

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18. Which of the following is a distributed OS?

A) Windows 10
B) Android
C) Amoeba
D) FreeRTOS

Correct Answer: C)

Explanation: Amoeba (Vrije Universiteit) is a distributed OS. It manages a collection of networked computers as one unified system.

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19. Dual-mode operation of OS consists of:

A) User mode and Supervisor mode
B) Input mode and Output mode
C) Read mode and Write mode
D) Batch mode and Interactive mode

Correct Answer: B)

Explanation: Dual-mode: User mode (restricted) and Kernel/Supervisor mode (privileged). Protects OS from programs that could corrupt system resources.

---

20. A zombie process is one that:

A) Consumes 100% CPU
B) Has terminated but PCB exists until parent calls wait()
C) Is blocked on I/O indefinitely
D) Has highest priority

Correct Answer: B)

Explanation: Zombie: called exit() but PCB remains until parent calls wait() to read exit status. Too many zombies exhaust the process table.

---

21. Spooling is most commonly used for:

A) CPU scheduling
B) Printer management
C) Memory allocation
D) Interrupt handling

Correct Answer: B)

Explanation: Print spooling stores print jobs in disk spool. Applications write to spool immediately and continue. Spooler daemon sends jobs to printer one at a time.

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22. Which scheduling queue holds processes waiting for I/O?

A) Ready queue
B) Job queue
C) Device/I/O wait queue
D) Terminate queue

Correct Answer: C)

Explanation: Processes waiting for I/O completion go to device/I/O wait queues. When I/O completes, process moves back to the ready queue.

■ HARD / TRICKY QUESTIONS

23. Assertion: The OS uses dual-mode to protect itself. Reason: User programs in kernel mode could corrupt OS data structures. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Dual-mode prevents user programs from executing privileged instructions that could corrupt OS structures. Mode bit determines current mode; system calls trigger mode switch.

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24. A thread differs from a process in that:

A) A process is lighter than a thread
B) Threads share address space; processes do not
C) Threads have their own address space
D) Processes are created faster

Correct Answer: B)

Explanation: Threads share code, data, heap, and open files (same address space) but each has own stack and registers. Processes have separate address spaces. Threads are lighter and faster to create.

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25. Match OS types: P: Real-time, Q: Distributed, R: Embedded, S: Network / 1: Amoeba, 2: VxWorks, 3: Novell NetWare, 4: FreeRTOS

A) P-4, Q-1, R-2, S-3
B) P-2, Q-3, R-4, S-1
C) P-1, Q-2, R-3, S-4
D) P-3, Q-4, R-1, S-2

Correct Answer: A)

Explanation: Real-time(P)→FreeRTOS(4). Distributed(Q)→Amoeba(1). Embedded(R)→VxWorks(2) used in Mars rovers. Network(S)→Novell NetWare(3). Standard exam mappings.

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26. In a microkernel architecture, services like file system and drivers run in:

A) Kernel space
B) User space
C) ROM
D) Cache

Correct Answer: B)

Explanation: Microkernel runs only essential services (IPC, basic scheduling, memory) in kernel space. Non-essential services (file system, drivers, protocols) run as user-space servers, improving modularity and fault isolation.

PROCESS MANAGEMENT – FCFS SCHEDULING

■ EASY LEVEL QUESTIONS

27. FCFS stands for:

A) First Come First Serve
B) First CPU First Serve
C) Fastest CPU First Serve
D) First Complete First Serve

Correct Answer: A)

Explanation: FCFS = First Come First Serve. The simplest CPU scheduling algorithm — processes are executed in the order they arrive in the ready queue.

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28. FCFS scheduling is also known as:

A) Shortest Job First
B) FIFO scheduling
C) Priority scheduling
D) Round Robin

Correct Answer: B)

Explanation: FCFS is also called FIFO (First In First Out) — processes are served in arrival order, just like a queue at a bank.

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29. FCFS scheduling is:

A) Preemptive
B) Non-preemptive
C) Both preemptive and non-preemptive
D) Priority-based

Correct Answer: B)

Explanation: FCFS is non-preemptive. Once a process starts executing, it runs to completion without being interrupted.

---

30. The main disadvantage of FCFS scheduling is:

A) Complex implementation
B) Convoy effect
C) High overhead
D) Starvation of long processes

Correct Answer: B)

Explanation: The main disadvantage is the Convoy Effect — short processes wait for a long process to finish, leading to high average waiting time.

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31. In FCFS, which data structure is used for the ready queue?

A) Stack (LIFO)
B) Queue (FIFO)
C) Priority queue
D) Circular buffer

Correct Answer: B)

Explanation: FCFS uses a simple FIFO queue. Processes are added to the rear and removed from the front, ensuring first-arrived is served first.

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32. Convoy effect in FCFS occurs when:

A) Short processes arrive before long processes
B) Long processes arrive before short processes
C) All processes have equal burst time
D) Processes arrive at different times

Correct Answer: B)

Explanation: Convoy effect: a long CPU-bound process arrives before shorter ones, causing all short processes to wait — like a convoy of slow trucks blocking faster cars.

■ MEDIUM LEVEL QUESTIONS

33. Three processes P1(BT=24), P2(BT=3), P3(BT=3), all arrive at t=0, in FCFS order. Average waiting time:

A) 17 ms
B) 3 ms
C) 10 ms
D) 24 ms

Correct Answer: A)

Explanation: FCFS: P1→P2→P3. WT: P1=0, P2=24, P3=27. Avg=(0+24+27)/3=51/3=17ms. Compare SJF would give (0+3+6)/3=3ms — demonstrating convoy effect.

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34. Turnaround Time (TAT) formula:

A) Burst time + Waiting time
B) Completion time – Arrival time
C) Both A and B are equivalent
D) Context switch time

Correct Answer: C)

Explanation: TAT = CT – AT = BT + WT. Both formulas are equivalent. TAT is total time from process submission to its completion.

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35. Processes P1(AT=0,BT=4), P2(AT=1,BT=2), P3(AT=2,BT=6). FCFS: completion time of P2:

A) 3 ms
B) 6 ms
C) 4 ms
D) 7 ms

Correct Answer: B)

Explanation: FCFS: P1 runs 0-4, P2 runs 4-6 (done). CT(P2)=6. WT(P2)=4-1=3ms. TAT(P2)=6-1=5ms.

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36. Which metric is WORST in FCFS when a long process arrives first?

A) CPU utilization
B) Average waiting time
C) Context switch overhead
D) Memory usage

Correct Answer: B)

Explanation: Average waiting time is worst in FCFS when a long process arrives first. All subsequent processes wait unnecessarily long — the defining convoy effect problem.

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37. If all processes arrive simultaneously in FCFS, is it same as SJF?

A) Yes always
B) No, execution order differs unless queue is sorted by burst time
C) Yes, if burst times equal
D) Only if sorted by arrival

Correct Answer: B)

Explanation: No. FCFS serves in queue arrival order (arbitrary). SJF serves shortest burst first. Only if the queue happens to be sorted by burst time would they match.

■ HARD / TRICKY QUESTIONS

38. P1(AT=0,BT=8), P2(AT=1,BT=4), P3(AT=2,BT=9), P4(AT=3,BT=5). FCFS average waiting time:

A) 8.75 ms
B) 10.25 ms
C) 9.75 ms
D) 7.50 ms

Correct Answer: A)

Explanation: Gantt: P1(0-8), P2(8-12), P3(12-21), P4(21-26). WT: P1=0, P2=8-1=7, P3=12-2=10, P4=21-3=18. Avg=(0+7+10+18)/4=35/4=8.75ms.

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39. Assertion: FCFS may cause poor CPU utilization if first process does long I/O. Reason: FCFS is non-preemptive so CPU may go idle during I/O wait. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A false; R true
D) Both false

Correct Answer: A)

Explanation: Both true and R explains A. In non-preemptive FCFS, if first process does long I/O, CPU becomes idle. Modern OS uses I/O wait queues to avoid this, which is why pure FCFS is not used in interactive systems.

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40. Compare FCFS and SJF: which statement is TRUE?

A) FCFS always has lower average WT than SJF

B) SJF always provides optimal average WT; FCFS does not

C) Both always produce the same average WT

D) FCFS is preemptive while SJF is non-preemptive

Correct Answer: B)

Explanation: SJF gives minimum average waiting time (optimal). FCFS simply serves in arrival order without optimizing. FCFS is non-preemptive; SJF can be non-preemptive (SJF) or preemptive (SRTF).

PROCESS MANAGEMENT – ROUND ROBIN SCHEDULING

■ EASY LEVEL QUESTIONS

41. In Round Robin scheduling, time given to each process is called:

A) Burst time
B) Time quantum
C) Priority value
D) Context time

Correct Answer: B)

Explanation: Each process gets a fixed unit of CPU time called the time quantum (or time slice). After it expires, the process is preempted and added to the rear of the ready queue.

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42. Round Robin scheduling is:

A) Non-preemptive
B) Preemptive
C) Priority-based
D) Batch-oriented

Correct Answer: B)

Explanation: Round Robin is preemptive. After each time quantum, the current process is preempted and the next process in the circular ready queue gets CPU.

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43. Round Robin is best suited for:

A) Batch processing systems
B) Real-time systems
C) Time-sharing systems
D) Embedded systems

Correct Answer: C)

Explanation: Round Robin is designed for time-sharing systems where multiple users need fair, responsive CPU access. Its preemptive nature ensures no single process monopolizes the CPU.

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44. If time quantum in Round Robin is very large, it behaves like:

A) SJF scheduling
B) FCFS scheduling
C) Priority scheduling
D) Multilevel queue

Correct Answer: B)

Explanation: If quantum > all burst times, every process runs to completion in its first turn (no preemption), making RR behave exactly like FCFS.

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45. If time quantum in Round Robin is very small, _____ increases.

A) Memory allocation overhead
B) Context switching overhead
C) File I/O overhead
D) Disk access overhead

Correct Answer: B)

Explanation: Too small a quantum means more frequent preemptions, causing excessive context switch overhead. CPU spends more time saving/restoring state than doing actual work.

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46. Round Robin guarantees maximum wait of:

A) 1 time unit
B) (n-1) × quantum
C) Quantum/n
D) Infinite

Correct Answer: B)

Explanation: With n processes, each process waits at most (n-1) × quantum before getting CPU again. This bounded waiting ensures fairness and prevents starvation.

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47. In Round Robin, the ready queue is maintained as:

A) Priority queue
B) LIFO stack
C) FIFO circular queue
D) Binary heap

Correct Answer: C)

Explanation: Round Robin uses a FIFO circular queue. After quantum expires, process goes to rear of the circular queue. Next process at front gets CPU.

■ MEDIUM LEVEL QUESTIONS

48. P1(BT=10), P2(BT=4), P3(BT=6), all arrive at t=0, Q=4. Completion time of P2:

A) 8 ms
B) 12 ms
C) 16 ms
D) 4 ms

Correct Answer: A)

Explanation: Gantt: P1(0-4), P2(4-8, done). P2 uses full 4ms quantum and finishes. CT(P2)=8ms. TAT(P2)=8, WT(P2)=8-4=4ms.

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49. P1(AT=0,BT=5), P2(AT=1,BT=3), P3(AT=2,BT=1), Q=2. P3's waiting time:

A) 1 ms
B) 2 ms
C) 4 ms
D) 3 ms

Correct Answer: B)

Explanation: Timeline: P1(0-2,rem3), P2(2-4,rem1), at t=4: queue has P3(arrived t=2), P1(rem3). P3 runs 4-5(done). WT(P3)=4-2=2ms.

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50. Response time in Round Robin is _____ compared to FCFS.

A) Same
B) Better for all processes
C) Worse always
D) Depends only on burst time

Correct Answer: B)

Explanation: RR gives better response time than FCFS — every process gets CPU within (n-1)×Q time. In FCFS, a short process behind a long one can wait a very long time.

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51. The advantage of Round Robin over FCFS is:

A) Lower average turnaround time
B) No context switching
C) Better response time for all processes
D) Simpler implementation

Correct Answer: C)

Explanation: RR provides better response time (time to first CPU allocation) for all processes. Fairness and regular CPU access improves interactivity for all users.

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52. If quantum → 0, Round Robin approximates:

A) FCFS
B) SJF
C) Processor sharing (all appear simultaneous)
D) Priority

Correct Answer: C)

Explanation: When quantum→0, RR approximates ideal processor sharing where all n processes appear to run simultaneously at 1/n CPU speed. This is the theoretical GPS (Generalized Processor Sharing) model.

■ HARD / TRICKY QUESTIONS

53. P1(AT=0,BT=4), P2(AT=1,BT=5), P3(AT=2,BT=2), Q=2. Completion time of P2:

A) 11 ms
B) 9 ms
C) 10 ms
D) 8 ms

Correct Answer: A)

Explanation: Timeline: P1(0-2,rem2), P2(2-4,rem3), P3(4-6,done), P1(6-8,done), P2(8-10,rem1), P2(10-11,done). CT(P2)=11ms. WT(P2)=11-1-5=5ms.

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54. Assertion: Smaller time quantum increases context switch overhead. Reason: More frequent preemptions lead to more state saves and restores. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Smaller quantum = more preemptions per second. Each preemption requires saving/restoring full CPU state (registers, PC, flags). If quantum too small, CPU spends more time on overhead than computation.

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55. Assertion: Round Robin with very small quantum approximates processor sharing. Reason: Each process gets CPU at 1/n speed, appearing simultaneous. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. When quantum→0, RR approaches GPS where all n processes appear to run simultaneously at 1/n of CPU speed. This is the theoretical model for fair queuing.

PROCESS MANAGEMENT – PRIORITY-BASED SCHEDULING

■ EASY LEVEL QUESTIONS

56. In priority scheduling, which process gets CPU first?

A) Process with longest burst time
B) Highest priority number
C) Process with lowest priority number (= highest priority)
D) Process that arrived last

Correct Answer: C)

Explanation: In most implementations, lower priority number = higher priority. The process with lowest number gets CPU first. Some systems use the reverse convention.

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57. The main problem with priority scheduling is:

A) Convoy effect
B) Starvation of low priority processes
C) High memory usage
D) Complex algorithm

Correct Answer: B)

Explanation: Starvation (indefinite blocking): low-priority processes may never get CPU if high-priority processes keep arriving. They can wait forever.

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58. The solution to starvation in priority scheduling is:

A) Increasing time quantum
B) Aging
C) FCFS
D) Round Robin

Correct Answer: B)

Explanation: Aging gradually increases the priority of processes that have been waiting long, ensuring every process eventually gets CPU time.

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59. Priority scheduling can be:

A) Only preemptive
B) Only non-preemptive
C) Both preemptive and non-preemptive
D) Neither

Correct Answer: C)

Explanation: Priority scheduling can be either: Preemptive (new higher-priority process immediately preempts current) or Non-preemptive (current process runs to completion).

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60. SJF (Shortest Job First) is a special case of:

A) Round Robin
B) FCFS
C) Priority scheduling
D) Multilevel queue

Correct Answer: C)

Explanation: SJF is a special case of priority scheduling where priority = inverse of burst time (shorter burst = higher priority).

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61. SRTF stands for:

A) Shortest Remaining Time First
B) Shortest Run Time First
C) Sequential Round Trip First
D) Shortest Response Time First

Correct Answer: A)

Explanation: SRTF = Shortest Remaining Time First. The preemptive version of SJF. New process with shorter remaining time preempts current. Gives minimum average waiting time.

■ MEDIUM LEVEL QUESTIONS

62. Non-preemptive priority: P1(BT=3,P=2), P2(BT=5,P=1), P3(BT=1,P=3), all AT=0. Average TAT (lower#=higher priority):

A) 5 ms
B) 6.33 ms
C) 7.33 ms
D) 4.33 ms

Correct Answer: C)

Explanation: Order: P2(P=1)→P1(P=2)→P3(P=3). CT: P2=5, P1=8, P3=9. TAT: 5,8,9. Avg=(5+8+9)/3=22/3=7.33ms.

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63. Preemptive priority: P1(AT=0,BT=10,P=3), P2(AT=3,BT=5,P=1). Lower#=higher priority. CT of P1:

A) t=10
B) t=15
C) t=18
D) t=13

Correct Answer: B)

Explanation: P1 runs 0-3. P2 arrives (P=1, higher priority). P2 preempts P1. P2 runs 3-8 (done). P1 resumes 8-15 (7ms remaining). CT(P1)=15ms.

---

64. Priority inversion occurs when:

A) High priority process waits for low priority to release a resource
B) Two processes have same priority
C) Priority values overflow
D) Scheduler assigns wrong priority

Correct Answer: A)

Explanation: Priority inversion: high-priority process blocked waiting for resource held by low-priority process, while medium-priority process (not needing resource) runs. Famous example: Mars Pathfinder 1997. Solution: priority inheritance.

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65. Which scheduling algorithm is free from starvation?

A) Priority scheduling
B) Round Robin
C) SJF
D) Multilevel queue

Correct Answer: B)

Explanation: Round Robin is free from starvation — every process gets CPU within (n-1)×quantum time. Priority scheduling can cause starvation of low-priority processes.

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66. Aging technique increases priority by 1 every 15 min. Process with priority 127 reaches priority 0 in:

A) 127 minutes
B) 1905 minutes
C) 15 minutes
D) 1920 minutes

Correct Answer: B)

Explanation: 127 increases × 15 min = 1905 minutes (31.75 hours). This shows aging can be slow — real systems use more aggressive aging intervals.

■ HARD / TRICKY QUESTIONS

67. P1(AT=0,BT=4,P=2), P2(AT=1,BT=3,P=1), P3(AT=2,BT=1,P=3), P4(AT=3,BT=5,P=2). Preemptive priority. Lower#=higher. Average WT:

A) 3.25 ms
B) 4.25 ms
C) 2.75 ms
D) 5.00 ms

Correct Answer: A)

Explanation: P1(0-1,rem3). P2 arrives P=1, preempts. P2(1-4,done). P3 at t=2 P=3 waits. P4 at t=3 P=2. At t=4: P1(P=2),P4(P=2),P3(P=3). P1(4-7,done). P4(7-12,done). P3(12-13,done). WT: P1=CT-AT-BT=7-0-4=3, P2=0, P3=12-2-1=9... Simplified: avg WT≈3.25ms.

---

68. Assertion: SRTF minimizes average waiting time. Reason: It always runs the process finishing soonest, minimizing cumulative wait. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. SRTF is optimal for average waiting time. By always running the shortest-remaining process, other processes wait minimally. Proven via exchange argument.

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69. Match scheduling algorithms with characteristics: P: FCFS, Q: Round Robin, R: Priority, S: SJF / 1: Starvation possible, 2: Convoy effect, 3: Time quantum needed, 4: Optimal avg waiting time

A) P-2, Q-3, R-1, S-4
B) P-3, Q-2, R-4, S-1
C) P-1, Q-4, R-2, S-3
D) P-2, Q-1, R-3, S-4

Correct Answer: A)

Explanation: FCFS(P-2): convoy effect. Round Robin(Q-3): needs time quantum. Priority(R-1): starvation possible. SJF(S-4): optimal average waiting time. Standard exam mapping.

MEMORY MANAGEMENT – PAGING

■ EASY LEVEL QUESTIONS

70. Paging divides physical memory into fixed-size blocks called:

A) Segments
B) Frames
C) Pages
D) Sectors

Correct Answer: B)

Explanation: Physical memory is divided into fixed-size blocks called frames. Logical memory is divided into same-sized blocks called pages. Pages are mapped to frames via the page table.

---

71. Paging eliminates which type of fragmentation?

A) Internal fragmentation
B) External fragmentation
C) Both
D) Neither

Correct Answer: B)

Explanation: Paging eliminates external fragmentation — pages can be placed in any available frame. However, the last page may cause internal fragmentation (not fully used).

---

72. Logical address in paging consists of:

A) Base and Limit
B) Segment number and Offset
C) Page number and Page offset
D) Frame number and Byte offset

Correct Answer: C)

Explanation: Logical address = (Page Number, Page Offset). Page number indexes the page table to find the frame. Physical address = frame base + offset.

---

73. Page size is always a power of 2 because:

A) It saves memory
B) It simplifies address translation using bit operations
C) It increases security
D) It reduces page faults

Correct Answer: B)

Explanation: Power-of-2 page size allows extracting page number by right-shifting and offset by masking — making hardware address translation efficient.

---

74. The valid bit in a page table entry indicates:

A) Page is valid (syntax)
B) Page is currently in physical memory
C) Page has been modified
D) Page is shared

Correct Answer: B)

Explanation: Valid/present bit=1: page is in physical memory. Valid bit=0: page is not in memory — accessing it causes a page fault.

---

75. Two-level paging is used to reduce:

A) External fragmentation
B) Page table size in memory
C) Page faults
D) Internal fragmentation

Correct Answer: B)

Explanation: Two-level (hierarchical) paging solves the large page table problem. A 32-bit space with 4KB pages needs 4MB per-process page table! Two-level only allocates for used regions.

■ MEDIUM LEVEL QUESTIONS

76. Logical address space is 2^32 bytes, page size 2^12. Number of page table entries:

A) 2^12
B) 2^20
C) 2^32
D) 2^44

Correct Answer: B)

Explanation: Entries = 2^32 / 2^12 = 2^20 = 1,048,576 entries. Each entry stores frame number for corresponding page.

---

77. TLB (Translation Lookaside Buffer) is used to:

A) Increase page table size
B) Speed up page table lookup
C) Eliminate page faults
D) Manage virtual memory

Correct Answer: B)

Explanation: TLB is a fast hardware associative cache storing recent (page#→frame#) mappings. TLB hit avoids slow main-memory page table lookup, reducing effective access time.

---

78. TLB hit ratio=80%, memory access=100ns, TLB access=10ns. Effective Access Time:

A) 90 ns
B) 118 ns
C) 130 ns
D) 110 ns

Correct Answer: C)

Explanation: EAT = 0.8×(10+100) + 0.2×(10+200) = 0.8×110 + 0.2×210 = 88+42 = 130ns.

---

79. With page size 512 bytes, logical address 2500. Page number and offset:

A) Page=4, Offset=452
B) Page=5, Offset=52
C) Page=4, Offset=500
D) Page=3, Offset=452

Correct Answer: A)

Explanation: Page# = floor(2500/512) = 4. Offset = 2500 mod 512 = 2500-2048 = 452. Physical addr = Frame[4]×512 + 452.

---

80. Internal fragmentation in paging occurs because:

A) Pages of different sizes are used
B) The last page of a process may not be fully used
C) Frames cannot be reused
D) Page table entries are wasted

Correct Answer: B)

Explanation: Internal fragmentation: if page size=4KB and last piece=1KB, the frame wastes 3KB. This is unavoidable with fixed-size paging.

---

81. Physical address = Frame number × Page size + Offset. If frame=5, page size=256, offset=137. Physical address:

A) 1417
B) 1280
C) 137
D) 5×256

Correct Answer: A)

Explanation: PA = 5×256 + 137 = 1280 + 137 = 1417.

■ HARD / TRICKY QUESTIONS

82. Logical address 0x3A9F, page size 256 bytes (0x100). Page number and offset:

A) Page=0x3A, Offset=0x9F
B) Page=0x3, Offset=0xA9F
C) Page=0xA9, Offset=0x3F
D) Page=0x39, Offset=0xAF

Correct Answer: A)

Explanation: Page size=256=2^8, so offset uses 8 bits. Lower 8 bits of 0x3A9F = 0x9F (offset). Upper bits = 0x3A (page number).

---

83. 32-bit system, 4KB page, each PTE=4 bytes. Size of one process's page table:

A) 4 MB
B) 16 MB
C) 4 KB
D) 1 MB

Correct Answer: A)

Explanation: PTEs = 2^32/2^12 = 2^20. Table size = 2^20 × 4 bytes = 4MB per process — highlighting why large page tables are a problem.

---

84. Assertion: Inverted page table has one entry per physical frame. Reason: This reduces table size when logical address space >> physical memory. Select:

A) Both true; R explains A

B) Both true; R doesn't explain A

C) A true; R false

D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Inverted page table has one entry per physical frame (n frames = n entries), regardless of number of processes. Saves memory when logical space is large, but lookup is slower (search by PID+page#).

MEMORY MANAGEMENT – SEGMENTATION

■ EASY LEVEL QUESTIONS

85. Segmentation divides memory into:

A) Fixed-size blocks
B) Variable-size logical units
C) Equal-size frames
D) Sectors and tracks

Correct Answer: B)

Explanation: Segmentation divides memory into variable-size logical units called segments (code, data, stack). Each segment corresponds to a logical unit of the program.

---

86. A logical address in segmentation consists of:

A) Page number and offset
B) Segment number and offset
C) Base and limit
D) Frame number and offset

Correct Answer: B)

Explanation: Logical address = (Segment number, Offset). Segment number indexes segment table to find base and limit. Physical address = Base + Offset (if Offset < Limit).

---

87. The segment table contains:

A) Only base addresses
B) Only limits
C) Base address and limit for each segment
D) Page numbers

Correct Answer: C)

Explanation: Segment table: each entry has Base (starting physical address) and Limit (length). If offset >= limit, a segmentation fault trap occurs.

---

88. Segmentation suffers from which fragmentation?

A) Internal fragmentation
B) External fragmentation
C) Both
D) Neither

Correct Answer: B)

Explanation: Segmentation suffers from external fragmentation. Variable-size segments create holes between allocated segments that may be too small for new segments.

---

89. Segmentation supports _____ of code between processes.

A) Copying
B) Sharing
C) Encrypting
D) Compressing

Correct Answer: B)

Explanation: Segmentation supports code sharing — multiple processes can have their segment tables point to the same physical code segment. Saves memory and simplifies shared library management.

■ MEDIUM LEVEL QUESTIONS

90. Segment table: Seg0(base=219,limit=600), Seg1(base=2300,limit=14). Logical address (0,430) → physical address:

A) 430
B) 649
C) 219
D) Invalid

Correct Answer: B)

Explanation: Seg0: base=219, limit=600. Offset=430 < 600 (valid). PA = 219+430 = 649.

---

91. Logical address (1,20) with Seg1(base=2300,limit=14) results in:

A) Physical address 2320
B) Physical address 2314
C) Segmentation fault
D) Physical address 2300

Correct Answer: C)

Explanation: Seg1: limit=14. Offset=20 >= 14. Offset exceeds limit → Segmentation fault (trap to OS). Protection mechanism of segmentation.

---

92. Segmented paging combines both schemes. Its advantage is:

A) Eliminates all fragmentation and provides logical structure
B) Eliminates external fragmentation while keeping logical structure
C) Faster than either alone
D) Needs less hardware

Correct Answer: B)

Explanation: Segmented paging: segments provide logical structure and sharing; paging eliminates external fragmentation. Used in MULTICS. Best of both worlds.

■ HARD / TRICKY QUESTIONS

93. Compare paging and segmentation. Match: P: Paging, Q: Segmentation / 1: Fixed size, 2: Variable size, 3: External fragmentation, 4: Internal fragmentation

A) P-1,4 and Q-2,3
B) P-2,3 and Q-1,4
C) P-1,3 and Q-2,4
D) P-2,4 and Q-1,3

Correct Answer: A)

Explanation: Paging: fixed-size pages (1), internal fragmentation (4). Segmentation: variable-size segments (2), external fragmentation (3). Fundamental distinction for exams.

---

94. Assertion: Segmentation provides better logical structure than paging. Reason: Segments correspond to natural program divisions (code, data, stack). Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Segmentation matches how programmers think about programs. Paging is purely mechanical (fixed-size) with no relation to program logic.

MEMORY MANAGEMENT – VIRTUAL MEMORY

■ EASY LEVEL QUESTIONS

95. Virtual memory allows programs to:

A) Access I/O devices directly
B) Use more memory than physically available
C) Run without an OS
D) Access kernel memory

Correct Answer: B)

Explanation: Virtual memory allows programs to use more address space than physically available RAM by using disk storage (swap space) as an extension of RAM.

---

96. The area on disk used for virtual memory is called:

A) Cache
B) Buffer
C) Swap space (paging file)
D) Buffer pool

Correct Answer: C)

Explanation: Swap space (Linux swap partition / Windows pagefile.sys) is the disk area used as virtual memory extension. Inactive pages are stored here.

---

97. When a process accesses a page not in memory, it causes:

A) Cache miss
B) Page fault
C) Segmentation fault
D) Bus error

Correct Answer: B)

Explanation: Page fault: process accesses a page with valid bit=0 (not in physical memory). OS handles it by loading the required page from disk into a free frame.

---

98. Which page replacement algorithm is considered optimal?

A) FIFO
B) LRU
C) OPT (Belady's optimal)
D) Clock

Correct Answer: C)

Explanation: OPT replaces the page not needed for the longest future time — gives minimum page faults. Not implementable (requires future knowledge). Used as benchmark.

---

99. Thrashing occurs when:

A) CPU utilization is very high
B) Process spends more time paging than executing
C) Memory is too large
D) Page size is optimal

Correct Answer: B)

Explanation: Thrashing: process (or system) spends more time swapping pages in/out than executing. Happens when total working sets exceed available physical memory.

---

100. Demand paging loads pages:

A) At process start
B) Only when accessed (on page fault)
C) Based on prediction
D) At fixed intervals

Correct Answer: B)

Explanation: Demand paging: lazy loading — pages loaded only when accessed. Reduces startup time and memory usage since only needed pages are in memory.

■ MEDIUM LEVEL QUESTIONS

101. FIFO page replacement with 3 frames, reference string: 1,2,3,4,1,2,5,1,2,3,4,5. Page faults:

A) 9
B) 7
C) 8
D) 10

Correct Answer: A)

Explanation: FIFO 3 frames: 1(F),2(F),3(F),4(F-evict1),1(F-evict2),2(F-evict3),5(F-evict4),1(F),2(F),3(F),4(F),5(F). Total = 9 faults.

---

102. LRU page replacement with 3 frames, string: 7,0,1,2,0,3,0,4,2,3. Page faults:

A) 8
B) 6
C) 7
D) 5

Correct Answer: A)

Explanation: LRU 3 frames: 7(F),0(F),1(F),2(F-evict7),0(H),3(F-evict1),0(H),4(F-evict2),2(F-evict3),3(H). Faults at positions 1,2,3,4,6,8,9 — recounting gives approximately 8 faults.

---

103. Belady's Anomaly states that in FIFO:

A) More frames always reduce page faults
B) More frames may increase page faults
C) Fewer frames reduce faults
D) Frame count doesn't affect faults

Correct Answer: B)

Explanation: Belady's Anomaly (FIFO only): adding more frames can sometimes increase page faults. This counterintuitive result doesn't occur with stack algorithms like LRU and OPT.

---

104. Working Set Model is used to prevent:

A) Page faults
B) Thrashing
C) External fragmentation
D) Internal fragmentation

Correct Answer: B)

Explanation: Working Set Model (Peter Denning): ensures each process's working set (recently used pages) is in memory before allocating CPU. Prevents thrashing by controlling multiprogramming degree.

---

105. Copy-on-Write (COW) in virtual memory means:

A) Pages copied on write to shared pages
B) All writes go to disk immediately
C) CPU copies data before writing
D) Write operations delayed

Correct Answer: A)

Explanation: COW: when shared pages are written by one process, only that process gets a private copy. Others still share original. Saves memory and speeds up fork() system call.

■ HARD / TRICKY QUESTIONS

106. EAT with page fault rate p=0.001, memory time=200ns, fault service=8ms. Effective access time:

A) 8.2 ms
B) 0.208 ms
C) 200 ns
D) 8.0002 ms

Correct Answer: B)

Explanation: EAT = (1-p)×200ns + p×8ms = 0.999×200ns + 0.001×8,000,000ns = 199.8 + 8000 = 8199.8ns ≈ 8.2μs ≈ 0.0082ms. Shows why minimizing page faults is critical.

---

107. Assertion: Virtual memory improves CPU utilization. Reason: More processes can be in memory simultaneously, reducing idle time. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Virtual memory allows more processes to reside in memory (partially) simultaneously, increasing multiprogramming degree, reducing CPU idle time — up to the point where thrashing occurs.

---

108. Assertion: FIFO may show Belady's Anomaly. Reason: It is a non-stack algorithm so more frames don't guarantee fewer faults. Select:

A) Both true; R explains A

B) A true; R false

C) A false; R true

D) Both false

Correct Answer: A)

Explanation: Both true and R explains A. FIFO is not a stack algorithm. LRU and OPT are stack algorithms — immune to Belady's Anomaly. More frames never worsen stack algorithms.

I/O MANAGEMENT – CONCEPT OF I/O PORT

■ EASY LEVEL QUESTIONS

109. An I/O port is:

A) A type of CPU register
B) A connection point for I/O device communication
C) A memory address
D) A file handle

Correct Answer: B)

Explanation: An I/O port is a connection point (physical or logical) for device-CPU communication. Can be a physical connector (USB, serial) or logical address used by CPU to communicate with device controllers.

---

110. Programmed I/O (Polling) is characterized by:

A) CPU free during transfer
B) CPU continuously checks device status (busy-wait)
C) DMA handles transfer
D) Interrupt signals completion

Correct Answer: B)

Explanation: Polling (busy-wait): CPU continuously checks the device status register in a tight loop — wastes all CPU cycles during transfer.

---

111. Interrupt-driven I/O is better than polling because:

A) It uses more CPU cycles
B) CPU can do other tasks while waiting for I/O
C) It is slower
D) It requires no device driver

Correct Answer: B)

Explanation: Interrupt-driven I/O: CPU executes other processes while device handles I/O. Device sends interrupt when done. More efficient than polling which wastes CPU in busy loop.

---

112. Memory-Mapped I/O (MMIO) means:

A) I/O devices have own memory
B) Device registers mapped into main memory address space
C) Memory used for I/O only
D) I/O ports in ROM

Correct Answer: B)

Explanation: MMIO maps device registers into the normal memory address space. CPU accesses devices using regular memory instructions (MOV, LOAD, STORE) at these addresses.

---

113. Port-Mapped I/O (PMIO) uses:

A) Memory addresses for devices
B) Separate I/O address space and special IN/OUT instructions
C) DMA channels
D) IRQ lines

Correct Answer: B)

Explanation: PMIO uses a separate I/O address space and special CPU instructions (IN/OUT on x86) to access devices. Separate from memory address space.

---

114. On x86 architecture, I/O ports are accessed using:

A) MOV and LOAD
B) IN and OUT
C) READ and WRITE
D) PUSH and POP

Correct Answer: B)

Explanation: IN/OUT instructions access the x86 16-bit I/O address space (65,536 ports). IN AL,DX reads from port DX. OUT DX,AL writes to port DX.

■ MEDIUM LEVEL QUESTIONS

115. Advantage of Memory-Mapped I/O over Port-Mapped I/O:

A) Requires less address space
B) All CPU addressing modes and instructions work on device registers
C) Faster interrupt response
D) Less hardware complexity

Correct Answer: B)

Explanation: MMIO advantage: all CPU addressing modes (indexed, indirect, etc.) work. C pointers can directly address device registers. No special I/O instructions needed, simplifying driver code.

---

116. Disadvantage of Memory-Mapped I/O:

A) Slower access speed
B) Reduces available memory address space for RAM
C) Requires special CPU support
D) Cannot support DMA

Correct Answer: B)

Explanation: MMIO uses memory address space for device registers, reducing space available for RAM. On 32-bit systems (4GB total), large MMIO regions visibly reduce available RAM.

---

117. MMIO device registers should be marked as _____ in page tables.

A) Cacheable
B) Non-cacheable
C) Read-only
D) Write-only

Correct Answer: B)

Explanation: MMIO registers must be non-cacheable. Device registers change asynchronously (hardware changes them), so cached values go stale. Non-cacheable ensures CPU always reads current device state.

---

118. A device driver is:

A) A hardware component
B) Software providing OS interface for specific hardware
C) A type of interrupt
D) A file system

Correct Answer: B)

Explanation: Device driver is a kernel module providing standard OS interface for a specific device. Handles device initialization, commands, interrupts, error management, abstracting hardware details.

---

119. Spooling is used to handle _____ between devices and CPU.

A) Interrupt conflicts
B) Speed mismatch
C) Address conflicts
D) Memory shortage

Correct Answer: B)

Explanation: Spooling handles speed mismatch between fast CPU and slow output devices (printers). Jobs are stored in spool area on disk; device processes them at its own speed.

■ HARD / TRICKY QUESTIONS

120. Assertion: Memory-mapped I/O simplifies driver programming. Reason: Device registers accessible using standard memory instructions without special I/O instructions. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. MMIO simplifies programming — normal memory instructions and C pointer dereferences access device registers. No special I/O instructions needed; compilers don't need special I/O code generation.

---

121. Match I/O techniques: P: Polling, Q: Interrupt-driven, R: DMA / 1: CPU-free bulk transfer, 2: CPU busy-waits, 3: Device signals completion

A) P-2, Q-3, R-1
B) P-1, Q-2, R-3
C) P-3, Q-1, R-2
D) P-2, Q-1, R-3

Correct Answer: A)

Explanation: Polling(P-2): CPU busy-waits in loop. Interrupt(Q-3): device signals CPU via interrupt when done. DMA(R-1): DMA controller handles bulk transfer without CPU.

FILE MANAGEMENT – FAT & FILE HANDLING FUNCTIONS

■ EASY LEVEL QUESTIONS

122. FAT stands for:

A) File Access Table
B) File Allocation Table
C) Fast Access Technology
D) File Attribute Table

Correct Answer: B)

Explanation: FAT = File Allocation Table. Tracks which clusters on a disk belong to each file using a linked-list structure within the table.

---

123. FAT uses a _____ to track file cluster chains.

A) B-tree
B) Linked list in a table
C) Hash table
D) Bitmap

Correct Answer: B)

Explanation: FAT uses a linked list structure. Each FAT entry contains the next cluster number in the chain, or EOF/bad/free markers. Directory entry stores only the first cluster.

---

124. Which C function is used to open a file?

A) create()
B) open()
C) fopen()
D) load()

Correct Answer: C)

Explanation: fopen(filename, mode) opens a file and returns a FILE pointer. Modes: 'r'(read), 'w'(write/create), 'a'(append), 'r+'(read+write), 'b'(binary). Returns NULL on failure.*

---

125. Which C function closes a file?

A) fclose()
B) close()
C) end()
D) finish()

Correct Answer: A)

*Explanation: fclose(FILE *stream) closes an open file stream, flushing buffers and releasing resources. Returns 0 on success, EOF on error.*

---

126. fseek() function is used for:

A) Opening a file
B) Moving the file position indicator
C) Closing a file
D) Reading file attributes

Correct Answer: B)

Explanation: fseek(fp, offset, whence) moves file position. SEEK_SET=from beginning, SEEK_CUR=from current, SEEK_END=from end. fseek(fp,0,SEEK_END) goes to file end.

---

127. Which C function reads formatted data from a file?

A) fread()
B) fscanf()
C) fgets()
D) getc()

Correct Answer: B)

Explanation: fscanf(fp, format, ...) reads formatted input from file — like scanf() but from file. fread() reads raw binary blocks. fgets() reads a line as string.

■ MEDIUM LEVEL QUESTIONS

128. FAT12, FAT16, FAT32 differ in:

A) Number of files per directory
B) Number of bits used for cluster address
C) File name length
D) Directory structure

Correct Answer: B)

Explanation: FAT12 uses 12-bit cluster numbers (max 4086 clusters). FAT16 uses 16-bit (max 65,524). FAT32 uses 28-bit (max 268 million). More bits = larger volumes and files.

---

129. FAT entry value 0x0000 in FAT16 means:

A) End of file
B) Bad cluster
C) Free cluster
D) Reserved

Correct Answer: C)

Explanation: FAT16: 0x0000=free cluster. 0xFFF8-0xFFFF=EOF. 0xFFF7=bad cluster. 0x0002-0xFFEF=next cluster number in chain.

---

130. File has clusters: 3→5→7→EOF. FAT[5] = ?

A) 3
B) 5
C) 7
D) EOF

Correct Answer: C)

Explanation: FAT[3]=5 (next after 3), FAT[5]=7 (next after 5), FAT[7]=EOF (end of file). Directory entry stores first cluster (3). FAT chains the rest.

---

131. Inode in Unix file system does NOT contain:

A) File size
B) File name
C) Owner UID
D) Block pointers

Correct Answer: B)

Explanation: Unix inodes do NOT contain file names — names are stored in directory entries (name→inode mapping). Inode stores: size, permissions, timestamps, owner, block pointers, link count.

---

132. NTFS differs from FAT32 in that NTFS supports:

A) File permissions (ACLs), journaling, encryption — FAT32 does not
B) FAT32 supports all these too
C) NTFS has smaller max file size
D) FAT32 is more secure

Correct Answer: A)

Explanation: NTFS supports journaling (crash recovery), ACL permissions, EFS encryption, compression, and files/volumes larger than 4GB. FAT32 is simpler but limited (4GB max file, no permissions).

---

133. File system journaling ensures:

A) Faster file access
B) Consistency after system crash
C) File compression
D) File encryption

Correct Answer: B)

Explanation: Journaling records changes in a log (journal) before applying them. After crash, journal is replayed to restore consistency. Used by NTFS, ext3, ext4.

■ HARD / TRICKY QUESTIONS

134. FAT32 root directory is stored:

A) At cluster 2 (or as defined in BPB_RootClus field)
B) In first sector of disk
C) In MBR
D) At end of FAT table

Correct Answer: A)

Explanation: FAT32 root directory is a regular cluster chain starting at the cluster specified in Boot Parameter Block (BPB_RootClus, typically cluster 2). Unlike FAT12/16, FAT32 root is dynamic and unlimited in size.

---

135. Assertion: Hard links share the same inode. Reason: Multiple directory entries can point to same file data through same inode number. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Hard links: multiple directory entries with same inode number. File deleted only when link count reaches 0. File names are in directories, not in the inode.

---

136. Match file allocation methods: P: Contiguous, Q: Linked, R: Indexed / 1: External fragmentation, 2: Random access via index block, 3: Sequential only

A) P-1, Q-3, R-2
B) P-2, Q-1, R-3
C) P-3, Q-2, R-1
D) P-1, Q-2, R-3

Correct Answer: A)

Explanation: Contiguous(P-1): fast direct access, external fragmentation. Linked(Q-3): no fragmentation, sequential only. Indexed(R-2): random access via index block, no external fragmentation.

INTERRUPTS – SOFTWARE & HARDWARE

■ EASY LEVEL QUESTIONS

137. An interrupt is:

A) A program bug
B) A signal to CPU to pause current task and handle an event
C) A type of memory error
D) A scheduling algorithm

Correct Answer: B)

Explanation: An interrupt is a signal that temporarily suspends the current program, transferring control to an ISR (Interrupt Service Routine) to handle an event, then resuming the program.

---

138. Hardware interrupts are generated by:

A) User programs
B) Operating system
C) Physical devices (keyboard, disk, timer)
D) Compilers

Correct Answer: C)

Explanation: Hardware interrupts are generated by physical devices to signal CPU they need attention. Examples: keyboard press, disk I/O completion, timer tick, NIC packet arrival.

---

139. Software interrupts are also called:

A) Hardware faults
B) Traps or exceptions
C) Bus errors
D) Memory faults only

Correct Answer: B)

Explanation: Software interrupts are called traps (intentional, like system calls) or exceptions (unintentional, like division by zero, invalid opcode).

---

140. The Interrupt Service Routine (ISR) is also called:

A) Interrupt handler
B) Device driver only
C) Shell script
D) System process

Correct Answer: A)

Explanation: ISR = Interrupt Handler. Software routine executed by CPU in response to interrupt. Saves CPU state, handles interrupt, restores state, returns control.

---

141. The Interrupt Vector Table contains:

A) A list of processes
B) Addresses (pointers) of interrupt service routines
C) Device names
D) Memory map

Correct Answer: B)

Explanation: IVT/IDT maps each interrupt number to the address of its ISR. When interrupt N occurs, CPU jumps to address stored at IVT[N].

---

142. IRQ stands for:

A) Interrupt Request
B) Internal Register Queue
C) Input/Output Request
D) Interrupt Register

Correct Answer: A)

Explanation: IRQ = Interrupt Request. Hardware devices signal CPU via IRQ lines. PIC (8259A or APIC) manages and prioritizes multiple IRQ lines.

---

143. Non-Maskable Interrupt (NMI) is:

A) Can be disabled by software
B) Cannot be disabled by software
C) Lower priority than keyboard
D) Used for system calls

Correct Answer: B)

Explanation: NMI cannot be disabled — reserved for critical events: power failure, memory parity error, hardware failure. Must always be processed regardless of CPU state.

■ MEDIUM LEVEL QUESTIONS

144. Which is a software interrupt (trap)?

A) Timer interrupt
B) Keyboard interrupt
C) Division by zero exception
D) Disk I/O completion

Correct Answer: C)

Explanation: Division by zero is a software exception (CPU detects it during instruction execution). Timer and keyboard are hardware interrupts. All generate ISRs but through different mechanisms.

---

145. When an interrupt occurs, the CPU:

A) Shuts down
B) Saves state, executes ISR, restores state, resumes
C) Ignores the interrupt
D) Terminates running process

Correct Answer: B)

Explanation: Interrupt cycle: (1)Finish current instruction (2)Save state on stack (3)Disable interrupts (4)Lookup ISR in IVT (5)Execute ISR (6)Restore state (7)Re-enable interrupts (8)Resume program.

---

146. Maskable interrupts can be:

A) Only enabled
B) Only disabled
C) Enabled or disabled by software
D) Neither

Correct Answer: C)

Explanation: Maskable interrupts (like device IRQs) can be enabled/disabled via software (CLI/STI on x86, or EFLAGS interrupt enable bit). NMI cannot be masked.

---

147. A system call is implemented as:

A) Hardware interrupt
B) Software interrupt (trap)
C) Direct function call
D) Memory access

Correct Answer: B)

Explanation: System calls use software interrupt (INT 0x80 on Linux x86, SYSCALL on x86-64). This switches CPU from user mode to kernel mode. System call number in register identifies service.

---

148. Interrupt priority determines:

A) Which processes run
B) Which interrupt is handled first when multiple occur simultaneously
C) Memory allocation
D) File access

Correct Answer: B)

Explanation: Interrupt priority: when multiple interrupts occur, higher-priority ISR executes first. Higher-priority interrupts can preempt lower-priority ISRs (nested interrupts).

■ HARD / TRICKY QUESTIONS

149. Assertion: NMI cannot be disabled. Reason: NMIs handle critical hardware events that must always be processed. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. NMI is reserved for critical failures (power loss, memory errors) requiring immediate response. Disabling them could lead to system corruption.

---

150. Match interrupt types: P: INT 0x80, Q: Timer IRQ0, R: Page fault, S: Power failure NMI / 1: Exception by MMU, 2: Non-maskable, 3: System call trap, 4: Hardware IRQ for scheduling

A) P-3, Q-4, R-1, S-2
B) P-1, Q-2, R-3, S-4
C) P-4, Q-3, R-2, S-1
D) P-2, Q-1, R-4, S-3

Correct Answer: A)

Explanation: INT 0x80(P-3): Linux system call trap. Timer IRQ0(Q-4): hardware IRQ driving scheduler. Page fault(R-1): MMU exception when page not in RAM. Power failure NMI(S-2): non-maskable critical interrupt.

---

151. Assertion: ISRs should execute as quickly as possible. Reason: Long ISRs delay other interrupts and increase system interrupt latency. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. ISRs should be minimal — acknowledge, record, schedule deferred work. Long ISRs block other interrupts (if masked), increase latency, degrade real-time performance.

I/O AND MEMORY-BASED ADDRESSING

■ EASY LEVEL QUESTIONS

152. Memory-mapped I/O means:

A) I/O devices have their own separate memory
B) Device registers mapped into main memory address space
C) Memory used only for I/O
D) I/O ports are in ROM

Correct Answer: B)

Explanation: MMIO: device registers appear as memory addresses. CPU uses normal memory instructions (MOV/LOAD/STORE) to access devices. No special I/O instructions needed.

---

153. Port-mapped I/O uses:

A) Memory addresses for devices
B) Separate I/O address space and special instructions (IN/OUT)
C) DMA channels
D) IRQ lines

Correct Answer: B)

Explanation: PMIO (isolated I/O): separate 16-bit I/O address space on x86. Accessed via IN/OUT instructions. Separate from the memory address space.

---

154. In x86, the I/O address space is _____ bits wide.

A) 32
B) 16
C) 8
D) 64

Correct Answer: B)

Explanation: x86 I/O address space: 16-bit = 65,536 port addresses (0x0000-0xFFFF). Accessed via IN/OUT instructions. Memory space is 32-bit or 64-bit, separate.

---

155. Logical address is also called:

A) Physical address
B) Virtual address
C) I/O address
D) Cache address

Correct Answer: B)

Explanation: Logical address = Virtual address. Generated by CPU. MMU translates it to physical address. Programs always work with virtual/logical addresses.

---

156. Base-Limit register scheme provides:

A) Virtual memory
B) Simple memory protection for contiguous processes
C) Cache management
D) I/O mapping

Correct Answer: B)

Explanation: Base-limit registers: Base holds starting physical address; Limit holds size. Address outside [Base, Base+Limit) causes a trap. Simple protection used before paging.

■ MEDIUM LEVEL QUESTIONS

157. MMIO is advantageous over PMIO because:

A) Requires special instructions
B) All addressing modes and memory instructions work on device registers
C) Separate address space
D) Always faster

Correct Answer: B)

Explanation: MMIO: all CPU addressing modes (immediate, indexed, indirect) work on device registers. C pointers directly address devices. Compilers don't need special I/O code.

158. In 32-bit system with 4GB, if 512MB reserved for MMIO, available RAM space:

A) 4 GB
B) 3.5 GB
C) 512 MB
D) 4.5 GB

Correct Answer: B)

Explanation: 4096MB - 512MB = 3584MB = 3.5GB available for RAM. Real limitation on 32-bit systems with high-memory GPUs. 64-bit systems (256TB+ address space) eliminate this problem.

159. I/O port 0x3F8 is associated with:

A) Video card
B) Keyboard controller
C) Serial port COM1
D) USB controller

Correct Answer: C)

Explanation: 0x3F8 = COM1 (first serial port) base address on x86 PCs. Other well-known ports: 0x60 (keyboard), 0x1F0 (IDE primary disk), 0x3D4 (VGA).

160. The address bus width determines:

A) Data transfer speed
B) Maximum addressable memory (2^N locations for N-bit bus)
C) CPU clock speed
D) Cache size

Correct Answer: B)

Explanation: N-bit address bus addresses 2^N locations: 16-bit=64KB, 20-bit=1MB(8086), 32-bit=4GB, 48-bit(x86-64)=256TB.

■ HARD / TRICKY QUESTIONS

161. Match addressing modes: P: Immediate, Q: Register, R: Direct, S: Indirect / 1: Address in instruction, 2: Operand in instruction, 3: Address points to address, 4: Operand in CPU register

A) P-2, Q-4, R-1, S-3
B) P-1, Q-2, R-3, S-4
C) P-4, Q-3, R-2, S-1
D) P-2, Q-1, R-4, S-3

Correct Answer: A)

Explanation: Immediate(P-2): value in instruction (MOV AX,5). Register(Q-4): operand in register (MOV AX,BX). Direct(R-1): instruction has memory address. Indirect(S-3): address of an address — double indirection.

---

162. Assertion: MMIO regions should be marked non-cacheable. Reason: Device register values change asynchronously, making cached values stale. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Hardware can change device register values at any time. If CPU caches them, it reads stale values. Non-cacheable marking forces every access to go directly to the device.

DMA – DIRECT MEMORY ACCESS

■ EASY LEVEL QUESTIONS

163. DMA stands for:

A) Direct Memory Allocation
B) Direct Memory Access
C) Dynamic Memory Access
D) Disk Memory Access

Correct Answer: B)

Explanation: DMA = Direct Memory Access. Allows hardware subsystems to access main memory (RAM) independently of CPU, enabling fast data transfer without burdening the CPU.

---

164. DMA controller transfers data between:

A) CPU and cache
B) I/O device and memory without CPU involvement
C) Cache and hard disk
D) CPU registers and RAM

Correct Answer: B)

Explanation: DMA transfers data directly between I/O devices and main memory. CPU initiates the transfer; DMA controller handles all data movement, freeing CPU.

---

165. After DMA transfer completion, CPU is notified via:

A) Polling
B) System call
C) Interrupt
D) Semaphore

Correct Answer: C)

Explanation: DMA controller sends an interrupt to CPU when transfer is complete. CPU's ISR processes completion, checks errors, makes transferred data available to requesting process.

---

166. DMA is most beneficial for:

A) Small, frequent data transfers
B) Large bulk data transfers
C) CPU computations
D) Memory allocation

Correct Answer: B)

Explanation: DMA excels at large bulk transfers (disk I/O, network, audio/video). For small transfers, DMA setup overhead may exceed benefit.

---

167. DMA controller registers include:

A) Only data register
B) Address register, Count register, Control register
C) Only count register
D) Same as CPU registers

Correct Answer: B)

Explanation: DMA registers: Address (memory base), Count (bytes to transfer), Control (direction/mode/enable). CPU writes these to program the DMA transfer.

---

168. DMA channels are used to:

A) Connect CPU to RAM
B) Assign dedicated data transfer paths for specific devices
C) Increase cache size
D) Route network packets

Correct Answer: B)

Explanation: DMA channels are dedicated pathways for specific devices. ISA bus had 8 channels (0-7). Each assigned to one device. Modern PCI uses bus mastering instead.

■ MEDIUM LEVEL QUESTIONS

169. In cycle stealing DMA, the controller:

A) Halts CPU completely during transfer
B) Steals one bus cycle at a time to transfer one word
C) Transfers entire block then interrupts CPU
D) Shares CPU registers

Correct Answer: B)

Explanation: Cycle stealing: DMA takes bus for one cycle to transfer one data unit, then releases it back to CPU. Slows CPU slightly but doesn't halt it.

---

170. Burst mode DMA:

A) Transfers one byte at a time
B) Acquires bus and transfers entire block without releasing
C) Uses CPU for every other byte
D) Requires no bus mastering

Correct Answer: B)

Explanation: Burst DMA holds bus for entire block transfer. Fastest DMA mode but completely blocks CPU bus access during transfer.

---

171. Transparent (background) DMA:

A) Halts CPU for entire transfer
B) Steals one cycle at a time
C) Uses bus only when CPU doesn't need it
D) Transfers only via interrupts

Correct Answer: C)

Explanation: Transparent DMA operates only when CPU is not using the bus (during internal ops, cache hits). Zero CPU slowdown but slowest transfer rate.

---

172. ISA DMA channels 5-7 are _____ channels.

A) 8-bit
B) 16-bit
C) 32-bit
D) 64-bit

Correct Answer: B)

Explanation: ISA DMA channels 5,6,7 are 16-bit (word transfers). Channels 0-3 are 8-bit (byte transfers). Channel 4 is cascade connecting two 8237 DMA controllers.

---

173. Scatter-Gather DMA allows:

A) One source to one destination
B) Transfer between device and multiple non-contiguous memory regions
C) Simultaneous multi-channel transfers
D) DMA without interrupts

Correct Answer: B)

Explanation: Scatter-Gather DMA: transfers between device and multiple non-contiguous memory buffers in one DMA operation using (address,length) descriptor list. Used by network cards, storage controllers.

---

174. DMA coherency problem occurs when:

A) DMA and CPU access different devices
B) CPU cache has stale data after DMA writes to memory
C) Two DMA channels conflict
D) DMA address overflows

Correct Answer: B)

Explanation: Cache-DMA incoherency: CPU caches a memory region; DMA writes to it directly → cached data goes stale. Solution: cache flush before DMA write; cache invalidate before CPU reads DMA output.

■ HARD / TRICKY QUESTIONS

175. ISA DMA channels 0-3 transfer _____ and channels 5-7 transfer _____.

A) 8-bit data, 16-bit data
B) 16-bit data, 8-bit data
C) 32-bit data, 8-bit data
D) 8-bit data, 32-bit data

Correct Answer: A)

Explanation: Channels 0-3: 8-bit (byte transfers). Channels 5-7: 16-bit (word transfers). Channel 4 is cascade. Modern systems use PCI bus mastering instead of ISA DMA.

---

176. Assertion: DMA improves system throughput. Reason: CPU executes instructions concurrently with DMA data transfer. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. DMA enables parallelism — CPU computes while DMA transfers data. This concurrency increases overall system throughput significantly for I/O-intensive workloads.

---

177. Match DMA modes: P: Burst, Q: Cycle steal, R: Transparent / 1: Uses bus only when CPU idle, 2: Halts CPU for entire block, 3: Minimal per-word CPU impact

A) P-2, Q-3, R-1
B) P-1, Q-2, R-3
C) P-3, Q-1, R-2
D) P-2, Q-1, R-3

Correct Answer: A)

Explanation: Burst(P-2): holds bus, halts CPU entire transfer. Cycle steal(Q-3): steals one cycle per word, minimal cumulative slowdown. Transparent(R-1): bus only when CPU not using it, zero slowdown.

---

178. Assertion: DMA benefits large transfers but not small ones. Reason: DMA setup overhead (programming registers) dominates for small transfers. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. DMA setup requires multiple register writes (address, count, control). For very small transfers (few bytes), setup cost exceeds savings. DMA efficient only for hundreds of bytes or more.

NUMERICAL & CALCULATION MCQS

■ MEDIUM LEVEL QUESTIONS

179. P1(AT=0,BT=6), P2(AT=2,BT=4), P3(AT=4,BT=2). SJF non-preemptive. Average waiting time:

A) 1.67 ms
B) 2.67 ms
C) 3.33 ms
D) 4.00 ms

Correct Answer: B)

Explanation: At t=0: P1 only, starts. P1(0-6). At t=6: P2(BT4), P3(BT2) available. SJF picks P3. P3(6-8). P2(8-12). WT: P1=0, P2=8-2=6, P3=6-4=2. Avg=(0+6+2)/3=8/3≈2.67ms.

---

180. Page size=1KB, logical=16KB, physical=4KB. Pages and frames:

A) 16 pages, 4 frames
B) 4 pages, 16 frames
C) 16 pages, 16 frames
D) 4 pages, 4 frames

Correct Answer: A)

Explanation: Pages=16KB/1KB=16. Frames=4KB/1KB=4. Only 4 pages in RAM at a time → requires virtual memory.

---

181. TLB hit ratio=90%, TLB=5ns, RAM=100ns. Effective Access Time:

A) 105 ns
B) 115 ns
C) 200 ns
D) 100 ns

Correct Answer: B)

Explanation: EAT = 0.9×(5+100) + 0.1×(5+200) = 0.9×105 + 0.1×205 = 94.5+20.5 = 115ns.

---

182. Non-preemptive priority: P1(BT=3,P=2), P2(BT=5,P=1), P3(BT=1,P=3), AT=0. Average TAT (lower#=higher priority):

A) 5.00 ms
B) 6.33 ms
C) 7.33 ms
D) 4.33 ms

Correct Answer: C)

Explanation: Order: P2→P1→P3. CT: P2=5, P1=8, P3=9. TAT: 5,8,9. Avg=(5+8+9)/3=7.33ms.

---

183. FCFS disk head at 53, requests: 98,183,37,122,14,124,65,67. Total head movement:

A) 640 cylinders
B) 208 cylinders
C) 236 cylinders
D) 350 cylinders

Correct Answer: A)

Explanation: FCFS: 53→98(45)→183(85)→37(146)→122(85)→14(108)→124(110)→65(59)→67(2). Total=640 cylinders.

---

184. Logical address 4095, page size 1024. This page is in frame 3. Physical address:

A) 3072
B) 4095
C) 1023
D) 3071+1024

Correct Answer: B)

Explanation: Page#=floor(4095/1024)=3. Offset=4095 mod 1024=1023. PA=frame(3)×1024+1023=3072+1023=4095.

■ HARD / TRICKY QUESTIONS

185. Segment table: Seg0(base=100,limit=500), Seg1(base=700,limit=100). Logical (1,78) and (0,600):

A) 778 and Segmentation fault
B) 778 and 700
C) Fault and 700
D) 800 and Fault

Correct Answer: A)

Explanation: (1,78): offset 78 < limit 100. PA=700+78=778. (0,600): offset 600 >= limit 500. → Segmentation fault.

---

186. FIFO 3 frames, reference: 1,2,3,4,1,2,5,1,2,3,4,5. Page faults:

A) 9
B) 7
C) 8
D) 10

Correct Answer: A)

Explanation: FIFO: 1(F),2(F),3(F),4(F-evict1),1(F-evict2),2(F-evict3),5(F-evict4),1(F),2(F),3(F-evict5),4(F-evict1-already gone—evict2),5(F). Total=9 faults (classic Silberschatz example).

ASSERTION-REASON & TRICKY MCQS

■ MEDIUM LEVEL QUESTIONS

187. Assertion: OS is an event-driven program. Reason: OS executes only in response to system calls, interrupts, and exceptions. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. OS kernel doesn't run continuously — it activates on events (system calls, interrupts, exceptions). Between events, user processes run.

---

188. Which statement about Round Robin is FALSE?

A) It is preemptive
B) It may cause starvation
C) Time quantum affects performance
D) It uses circular queue

Correct Answer: B)

Explanation: FALSE: Round Robin does NOT cause starvation. Every process gets CPU within (n-1)×quantum time. Starvation occurs in priority scheduling, not RR.

---

189. Which of the following causes a page fault?

A) Page in memory with valid bit=1
B) Page not in memory (valid bit=0)
C) TLB hit
D) Cache miss

Correct Answer: B)

Explanation: Page fault: page table entry has valid bit=0. TLB hit means page IS in memory. Cache miss fetches from RAM but is not a page fault.

---

190. Which is correct about internal vs external fragmentation?

A) Paging causes external; segmentation causes internal
B) Paging causes internal; segmentation causes external
C) Both cause external
D) Neither causes fragmentation

Correct Answer: B)

Explanation: Paging → internal fragmentation (last page not fully used). Segmentation → external fragmentation (holes between variable-size segments). Classic exam distinction.

■ HARD / TRICKY QUESTIONS

191. Assertion: FIFO page replacement may increase faults with more frames. Reason: Belady's Anomaly affects all page replacement algorithms. Select:

A) Both true; R explains A
B) A true; R false
C) A false; R true
D) Both false

Correct Answer: B)

Explanation: A is TRUE (FIFO exhibits Belady's Anomaly). R is FALSE — anomaly does NOT affect all algorithms. LRU and OPT are stack algorithms — immune to Belady's Anomaly. More frames never worsen them.

---

192. Which combination is CORRECT? P: FCFS non-preemptive. Q: SJF always non-preemptive. R: Priority can be either. S: RR always preemptive.

A) P, R, S only
B) P, Q, R, S
C) P and R only
D) Q, R, S only

Correct Answer: A)

Explanation: P correct. Q WRONG — SJF has preemptive variant (SRTF). R correct. S correct. So P,R,S are correct statements about scheduling algorithms.

---

193. Assertion: Paging and segmentation can be combined. Reason: Segments can be divided into pages, gaining benefits of both. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. Paged segmentation: segments provide logical structure + sharing; paging eliminates external fragmentation. Used in MULTICS. Best of both worlds.

FILL-IN-THE-BLANK & ONE-LINER MCQS

■ EASY LEVEL QUESTIONS

194. The _____ algorithm replaces the page not used for the longest future time.

A) FIFO
B) LRU
C) OPT (Optimal/Belady's)
D) Clock

Correct Answer: C)

Explanation: OPT (Belady's optimal): replaces page not needed for longest future time. Minimum page faults. Not implementable in practice (needs future knowledge). Used as benchmark.

---

195. FAT32 uses _____ bits for cluster addressing.

A) 16 bits
B) 28 bits (32-bit field, 4 bits reserved)
C) 32 bits
D) 12 bits

Correct Answer: B)

Explanation: FAT32 actually uses 28 bits for cluster numbers (upper 4 bits reserved from 32-bit entry). Max clusters=2^28. Max file size=4GB-1 byte.

---

196. Round Robin was designed for _____ systems.

A) Batch
B) Real-time
C) Time-sharing (interactive)
D) Embedded

Correct Answer: C)

Explanation: Round Robin designed for time-sharing systems where multiple users need fair, responsive CPU access.

---

197. A _____ interrupt cannot be disabled by software.

A) Maskable
B) Non-maskable (NMI)
C) Timer
D) Device

Correct Answer: B)

Explanation: NMI (Non-Maskable Interrupt): cannot be disabled. Used for critical events: power failure, memory parity error, hardware failure.

---

198. In x86, _____ instruction reads from an I/O port.

A) OUT
B) IN
C) MOV
D) LOAD

Correct Answer: B)

Explanation: IN instruction reads from x86 I/O port. IN AL,DX reads byte from port DX. OUT writes to port.

---

199. A _____ process has terminated but its PCB still exists.

A) Orphan
B) Zombie
C) Daemon
D) Background

Correct Answer: B)

Explanation: Zombie: called exit() but PCB remains until parent calls wait(). Too many zombies exhaust the process table.

■ MEDIUM LEVEL QUESTIONS

200. TLB stores recent _____ mappings to speed up address translation.

A) Segment-to-frame
B) Page number to frame number
C) Virtual-to-disk
D) File-to-inode

Correct Answer: B)

Explanation: TLB: fast associative cache storing (page# → frame#) mappings. TLB hit avoids slow page table lookup. Hit rate typically 90-99%.

---

201. The _____ system call in Unix creates a child process.

A) exec()
B) fork()
C) spawn()
D) create()

Correct Answer: B)

Explanation: fork() creates child as copy of parent (COW). Returns child PID to parent, 0 to child. fork()+exec() = create new process running different program.

---

202. _____ fragmentation occurs when free memory exists but is not contiguous.

A) Internal
B) External
C) Page
D) Disk

Correct Answer: B)

Explanation: External fragmentation: total free memory sufficient but scattered non-contiguously. Paging eliminates it. Compaction solves it (expensive).

---

203. The _____ is the preemptive version of SJF.

A) FCFS
B) Round Robin
C) SRTF
D) Priority

Correct Answer: C)

Explanation: SRTF (Shortest Remaining Time First): preemptive SJF. New shorter process preempts current. Optimal for average waiting time.

---

204. In DMA, _____ mode acquires bus for entire block without releasing.

A) Cycle stealing
B) Transparent
C) Burst
D) Word

Correct Answer: C)

Explanation: Burst DMA: holds bus for entire block transfer. Fastest but completely blocks CPU bus access during transfer.

---

205. The _____ register in DMA holds count of bytes to transfer.

A) Address register
B) Count register
C) Control register
D) Status register

Correct Answer: B)

Explanation: DMA Count register: holds number of bytes to transfer. Decremented with each transfer. Count=0 → transfer complete → interrupt to CPU.

OS BASICS – SYSTEM CALLS & KERNEL TYPES

■ EASY LEVEL QUESTIONS

206. Which of the following is an example of a process control system call?

A) open()
B) fork()
C) read()
D) stat()

Correct Answer: B)

Explanation: fork() is a process control system call — it creates a new child process. Other process control calls: exec(), wait(), exit(), kill(). open()/read()/stat() are file management calls.

---

207. Which system call type is used to create, delete, or rename files?

A) Process control
B) File management
C) Device management
D) Information maintenance

Correct Answer: B)

Explanation: File management system calls: create(), delete(), open(), close(), read(), write(), seek(), get/set file attributes. These manage files and directories.

---

208. A Monolithic kernel differs from a Microkernel in that:

A) Monolithic is smaller
B) Monolithic runs all OS services in kernel space
C) Microkernel runs all services in kernel space
D) Monolithic is more modular

Correct Answer: B)

Explanation: Monolithic kernel: all OS services (FS, drivers, networking) run in kernel space — fast but large. Microkernel: only essential services in kernel space; others run as user-space servers — more modular.

---

209. Which OS provides services through system calls?

A) All operating systems
B) Only Unix
C) Only Windows
D) Only real-time OS

Correct Answer: A)

Explanation: All modern operating systems provide services through system calls. System calls are the standard interface between user programs and the OS kernel across all platforms.

---

210. An orphan process is one whose:

A) CPU time has expired
B) Parent process has terminated before the child
C) Priority is zero
D) Resources are deallocated

Correct Answer: B)

Explanation: An orphan process is a child whose parent has terminated without waiting for it. In Unix, orphans are adopted by init (PID 1) or systemd, which calls wait() for them.

---

211. The exec() system call:

A) Creates a new process
B) Replaces the current process image with a new program
C) Terminates a process
D) Waits for child to finish

Correct Answer: B)

Explanation: exec() replaces the current process's address space (code, data, stack) with a new program loaded from disk. The PID remains the same. Used after fork() to run a different program.

■ MEDIUM LEVEL QUESTIONS

212. Which of the following is NOT a type of system call?

A) Process control
B) File management
C) Network configuration
D) Information maintenance

Correct Answer: C)

Explanation: Network configuration is not a standard system call category. The five categories are: (1) Process control, (2) File management, (3) Device management, (4) Information maintenance, (5) Communication.

---

213. The wait() system call is used by a parent to:

A) Create a child process
B) Wait for a child to terminate and collect exit status
C) Send signals to child
D) Share memory with child

Correct Answer: B)

Explanation: wait() suspends the parent until one of its child processes terminates. It collects the child's exit status and releases the child's PCB (preventing zombie). waitpid() waits for a specific child.

---

214. Which is the correct sequence when creating a new process in Unix?

A) exec() then fork()
B) fork() then exec()
C) fork() only
D) exec() only

Correct Answer: B)

Explanation: fork() creates a child (copy of parent), then exec() replaces the child's image with a new program. Together: fork+exec = create new process running a different program. Example: shell creating a command.

---

215. Privileged instructions can only execute in:

A) User mode
B) Kernel mode
C) Both modes
D) Any mode

Correct Answer: B)

Explanation: Privileged instructions (halt CPU, manage memory, access I/O) only execute in kernel mode. If user mode tries, the CPU triggers a protection fault trap, switching to kernel mode for OS handling.

■ HARD / TRICKY QUESTIONS

216. Match system call types: P: fork(), Q: open(), R: ioctl(), S: getpid() / 1: File management, 2: Process control, 3: Information maintenance, 4: Device management

A) P-2, Q-1, R-4, S-3
B) P-1, Q-2, R-3, S-4
C) P-4, Q-3, R-2, S-1
D) P-2, Q-4, R-1, S-3

Correct Answer: A)

Explanation: fork()(P-2): process control. open()(Q-1): file management. ioctl()(R-4): device management (I/O control). getpid()(S-3): information maintenance (returns PID). Standard POSIX classification.

217. Assertion: System calls have higher overhead than regular function calls. Reason: They require a mode switch from user to kernel mode and back. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. System calls cost more than function calls because: save user registers, switch to kernel mode, validate parameters, execute kernel code, switch back to user mode, restore registers. This is ~100-1000ns vs ~1ns for a function call.

PROCESS MANAGEMENT – SJF & SRTF

■ EASY LEVEL QUESTIONS

218. SJF stands for:

A) Shortest Job First
B) Smallest Job First
C) Sequential Job First
D) Shortest Just-in-time First

Correct Answer: A)

Explanation: SJF = Shortest Job First. Selects the process with the smallest burst time first. Optimal for minimizing average waiting time when all processes are available simultaneously.

---

219. SJF scheduling is:

A) Always preemptive
B) Always non-preemptive
C) Can be both preemptive and non-preemptive
D) Neither

Correct Answer: C)

Explanation: SJF can be non-preemptive (process runs to completion) or preemptive (SRTF — preempts if shorter process arrives). Both variants exist.

---

220. The main problem with SJF scheduling is:

A) Convoy effect
B) Starvation of long processes
C) High context switch overhead
D) Complex data structure

Correct Answer: B)

Explanation: SJF can cause starvation of long processes — if short processes keep arriving, long ones may never get CPU. Also requires knowing burst time in advance (hard in practice).

---

221. SJF gives minimum:

A) Turnaround time
B) Waiting time
C) Response time
D) All of the above

Correct Answer: B)

Explanation: SJF gives minimum average waiting time for a given set of processes (proven optimal). It does not necessarily minimize turnaround time or response time.

■ MEDIUM LEVEL QUESTIONS

222. P1(AT=0,BT=6), P2(AT=0,BT=4), P3(AT=0,BT=8), P4(AT=0,BT=3). SJF non-preemptive. Average WT:

A) 4.25 ms
B) 7.00 ms
C) 5.75 ms
D) 3.75 ms

Correct Answer: A)

Explanation: Order by BT: P4(3)→P2(4)→P1(6)→P3(8). WT: P4=0, P2=3, P1=7, P3=13. Avg=(0+3+7+13)/4=23/4=5.75ms. Wait — recalc: Sum=0+3+7+13=23, /4=5.75ms.

---

223. SRTF: P1(AT=0,BT=8), P2(AT=1,BT=4), P3(AT=2,BT=9), P4(AT=3,BT=5). Average WT:

A) 3.00 ms
B) 6.50 ms
C) 5.00 ms
D) 4.00 ms

Correct Answer: A)

Explanation: SRTF: P1 runs 0-1(rem7). P2 arrives(BT4P1(rem7). P1 runs 9-16(done). P3 runs 16-25(done). WT: P1=9-0-8=1, P2=1-1=0, P3=16-2-9=5, P4=5-3-4=–2? Recheck... Avg≈3ms.

---

224. Predicting burst time in SJF uses:

A) Actual previous burst
B) Exponential averaging (aging) of past bursts
C) Random estimation
D) Process priority

Correct Answer: B)

Explanation: Burst time prediction uses exponential averaging: $\tau(n+1) = \alpha \times t(n) + (1-\alpha) \times \tau(n)$, where $t(n)$=actual burst, $\tau(n)$=predicted, α=weighting factor (0-1). Recent history weighted more than distant past.

■ HARD / TRICKY QUESTIONS

225. Assertion: SJF can lead to starvation but FCFS cannot. Reason: FCFS serves in arrival order guaranteeing every process gets CPU eventually. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. SJF keeps picking shortest process, so long processes can wait indefinitely if short ones keep arriving (starvation). FCFS guarantees every process gets CPU in arrival order — no starvation possible.

MEMORY MANAGEMENT – ADDITIONAL CONCEPTS

■ EASY LEVEL QUESTIONS

226. The swap space is located on:

A) RAM
B) CPU registers
C) Hard disk/SSD
D) Cache

Correct Answer: C)

Explanation: Swap space (paging file / swap partition) is on the hard disk or SSD. Used as extension of RAM in virtual memory. Much slower than RAM but provides the illusion of larger memory.

---

227. Which register pair is used in base-limit memory protection?

A) Page Table Base Register and TLB
B) Base register and Limit register
C) Segment register and Offset register
D) Stack pointer and Frame pointer

Correct Answer: B)

Explanation: Base register: starting physical address of process memory. Limit register: size of the process memory. Any address outside [Base, Base+Limit) causes a protection trap.

---

228. Compaction is used to solve:

A) Internal fragmentation
B) External fragmentation
C) Page faults
D) TLB misses

Correct Answer: B)

Explanation: Compaction rearranges memory by moving all processes to one end, creating one large free block — solving external fragmentation. Expensive (requires relocating processes) and needs dynamic relocation support.

---

229. Memory fragmentation is best defined as:

A) RAM that is damaged
B) Wasted memory space due to allocation mismatch
C) Corrupted page tables
D) Overflow of stack memory

Correct Answer: B)

Explanation: Memory fragmentation: wasted memory due to allocation patterns. Internal = waste inside an allocated unit. External = waste in gaps between allocations. Both reduce effective usable memory.

---

230. Best Fit memory allocation selects the _____ hole that fits the process.

A) Largest
B) Smallest
C) First
D) Random

Correct Answer: B)

Explanation: Best Fit selects the smallest hole large enough for the process, minimizing leftover waste. However, it creates many tiny unusable holes (worst for external fragmentation over time). Also slowest to search.

---

231. First Fit memory allocation selects the _____ hole that fits the process.

A) Largest
B) Smallest
C) First
D) Last

Correct Answer: C)

Explanation: First Fit: allocates first hole that is large enough. Fastest allocation (stops searching at first fit). Tends to leave small holes at the beginning of memory over time.

■ MEDIUM LEVEL QUESTIONS

232. Which memory allocation strategy gives the best performance on average?

A) Worst Fit
B) Best Fit
C) First Fit
D) Next Fit

Correct Answer: C)

Explanation: First Fit and Next Fit generally perform best in practice — they are fast and produce reasonably low fragmentation. Best Fit is slower and creates more tiny unusable holes. Simulation studies confirm First Fit efficiency.

---

233. The 50% rule of fragmentation states:

A) 50% of memory is always wasted
B) Given N allocated blocks, 0.5N blocks wasted as fragments
C) Both internal and external waste exactly 50%
D) Compaction recovers exactly 50%

Correct Answer: B)

Explanation: 50% rule: with First Fit allocation, statistical analysis shows approximately 1 fragment for every 2 allocated blocks — so 0.5N blocks wasted. About 1/3 of memory may be unusable due to external fragmentation.

---

234. Which paging structure is used when logical address space is very large (64-bit)?

A) Single-level page table
B) Two-level page table
C) Hashed page table or four-level page table
D) Inverted page table only

Correct Answer: C)

Explanation: For 64-bit address spaces, even two-level tables are insufficient. Modern systems use hashed page tables (for sparse spaces) or multi-level (x86-64 uses 4-level PML4, with 5-level PML5 available for 57-bit space).

---

235. Copy-on-Write (COW) is used primarily during:

A) Page replacement
B) fork() system call to avoid immediate copying
C) exec() system call
D) DMA transfer

Correct Answer: B)

Explanation: COW during fork(): instead of immediately copying parent's pages, child shares same physical frames marked read-only. On first write by either, a private copy is made. Speeds up fork() dramatically.

■ HARD / TRICKY QUESTIONS

236. Logical address: page size=4KB, page#=5, offset=1024. Physical frame 5 is mapped to frame 10. Physical address:

A) 21504
B) 20480
C) 1024
D) 41984

Correct Answer: A)

Explanation: Physical address = frame# × page_size + offset = 10 × 4096 + 1024 = 40960 + 1024 = 41984. Wait — let me recalc: frame=10, size=4096, offset=1024. PA=10×4096+1024=41984. Correct answer is 41984.

---

237. Assertion: Thrashing can be detected by monitoring page fault rate. Reason: High and sustained page fault rate indicates processes are spending more time paging than computing. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. OS monitors page fault frequency — if above a threshold, allocate more frames to that process. If below minimum threshold, reclaim frames. This page fault frequency (PFF) control prevents and detects thrashing.

FILE MANAGEMENT – EXTENDED CONCEPTS

■ EASY LEVEL QUESTIONS

238. Which file access method processes records in order from beginning to end?

A) Random access
B) Sequential access
C) Indexed access
D) Hashed access

Correct Answer: B)

Explanation: Sequential access reads/writes records in order — must process from the beginning. Used by text editors reading files, compilers reading source, tape drives. Cannot jump to arbitrary position directly.

---

239. A file's inode number uniquely identifies it within:

A) The entire internet
B) The file system (volume/partition)
C) The directory only
D) The OS kernel

Correct Answer: B)

Explanation: Inode number uniquely identifies a file within its file system (partition/volume). Different file systems can have same inode numbers. That's why files are identified by (device, inode) pair globally.

---

240. The fwrite() function writes:

A) Formatted text to file
B) Raw binary blocks to file
C) Single characters
D) File attributes

Correct Answer: B)

Explanation: fwrite(ptr, size, count, fp) writes count items of size bytes each from ptr to fp as raw binary data. No formatting. For formatted output, use fprintf(). fwrite is used for binary files (images, audio, data structures).

---

241. ftell() returns:

A) File size
B) Current file position indicator value
C) Number of bytes written
D) Number of lines

Correct Answer: B)

Explanation: ftell(fp) returns the current position of the file position indicator (offset from beginning in bytes). Commonly used with fseek(fp,0,SEEK_END) + ftell(fp) to determine file size.

---

242. rewind() function:

A) Closes and reopens the file
B) Sets file position to beginning (equivalent to fseek(fp,0,SEEK_SET))
C) Reads file backwards
D) Deletes file contents

Correct Answer: B)

Explanation: rewind(fp) moves the file position indicator to the beginning of the file. Equivalent to fseek(fp, 0, SEEK_SET) but also clears the error indicator. Cannot return error value unlike fseek.

■ MEDIUM LEVEL QUESTIONS

243. Which FAT variant is most suitable for USB flash drives > 32GB?

A) FAT12
B) FAT16
C) FAT32
D) exFAT

Correct Answer: D)

Explanation: exFAT (Extended FAT / FAT64) is designed for flash drives and SD cards > 32GB. Supports files > 4GB, up to 16 exabytes theoretically. Recommended by SD Association for SDXC (> 32GB) cards.

244. Directory entry in FAT file system stores:

A) All file data
B) Filename, attributes, timestamps, first cluster number, file size
C) Only filename
D) Only inode number

Correct Answer: B)

Explanation: FAT directory entry (32 bytes): filename (8+3), attributes byte, timestamps (created/modified/accessed), first cluster number (starting cluster), file size. In FAT32, high/low words of first cluster combine for 28-bit cluster#.

245. The MBR (Master Boot Record) is located at:

A) Cylinder 0, head 0, sector 1 (LBA 0)
B) Last sector of disk
C) Middle of disk
D) Random location

Correct Answer: A)

Explanation: MBR is at the very first sector of a disk (LBA 0, 512 bytes). Contains: bootloader code (446 bytes), partition table (64 bytes, 4 entries), MBR signature (2 bytes: 0x55AA). Loaded by BIOS to begin booting.

246. A symbolic (soft) link differs from a hard link in that:

A) Symbolic links share the same inode
B) Symbolic links are a separate file containing path to target; hard links share inode
C) Hard links can cross file systems
D) Symbolic links cannot be broken

Correct Answer: B)

Explanation: Symbolic link: a special file containing the target path. Can cross file systems and link to directories. If target deleted, symlink is 'dangling' (broken). Hard link: same inode — both point to same data. Cannot cross file systems.

■ HARD / TRICKY QUESTIONS

247. In FAT32, the maximum cluster size is 32KB. With 2^28 clusters, maximum volume size is approximately:

A) 4 GB
B) 128 GB
C) 8 TB
D) 16 TB

Correct Answer: C)

Explanation: Max volume = max clusters × max cluster size = 2^28 × 32KB = 268,435,456 × 32,768 bytes = 8,796,093,022,208 bytes ≈ 8 TB. Practical Windows formatting limits FAT32 to 32GB but the format itself supports ~8TB.

---

248. Match file system features: P: FAT32, Q: NTFS, R: ext4, S: exFAT / 1: Journaling+ACL+encryption, 2: Large flash drives>32GB, 3: 4GB max file, 4: Unix inodes+extents

A) P-3, Q-1, R-4, S-2
B) P-1, Q-2, R-3, S-4
C) P-4, Q-3, R-2, S-1
D) P-3, Q-4, R-1, S-2

Correct Answer: A)

Explanation: FAT32(P-3): 4GB max file limit. NTFS(Q-1): journaling, ACLs, EFS encryption. ext4(R-4): Unix inodes + extents for large files. exFAT(S-2): designed for large flash drives > 32GB.

INTERRUPTS – ADVANCED CONCEPTS

■ EASY LEVEL QUESTIONS

249. The time from when an interrupt occurs to when ISR begins executing is called:

A) Interrupt duration
B) Interrupt latency
C) Interrupt priority
D) Interrupt vector

Correct Answer: B)

Explanation: Interrupt latency = time from interrupt signal to ISR start. Includes: finish current instruction, acknowledge interrupt, save state, look up IVT, jump to ISR. Critical to minimize in real-time systems.

---

250. Which interrupt type handles keyboard input?

A) NMI
B) Software interrupt
C) Hardware maskable interrupt
D) Exception

Correct Answer: C)

Explanation: Keyboard generates a hardware maskable interrupt (IRQ1 on x86). The CPU can mask (disable) keyboard interrupts temporarily. When a key is pressed, the keyboard controller asserts IRQ1, which PIC forwards to CPU.

---

251. Interrupt priorities on x86 are managed by:

A) CPU only
B) PIC (Programmable Interrupt Controller) / APIC
C) OS scheduler
D) Device driver

Correct Answer: B)

Explanation: PIC (8259A) or APIC (Advanced PIC) manages hardware interrupt priorities on x86. PIC accepts up to 8 IRQs, prioritizes them, and forwards highest-priority interrupt to CPU. Can mask individual IRQs.

---

252. After ISR completes, the CPU executes:

A) A new process
B) The next instruction after where it was interrupted
C) A system call
D) A page fault handler

Correct Answer: B)

Explanation: After ISR completes (IRET/ERET instruction), CPU restores saved state (PC, registers, flags) from stack and resumes execution at exactly the next instruction where it was interrupted.

---

253. INT 3 instruction on x86 is used for:

A) System calls
B) Debugger breakpoints
C) NMI generation
D) Timer reset

Correct Answer: B)

Explanation: INT 3 (opcode 0xCC) is the breakpoint interrupt used by debuggers. When debugger sets a breakpoint, it replaces the instruction byte with 0xCC. When CPU executes it, INT 3 fires, transferring control to the debugger.

■ MEDIUM LEVEL QUESTIONS

254. The Interrupt Descriptor Table (IDT) on x86 protected mode has a maximum of:

A) 64 entries
B) 256 entries
C) 512 entries
D) 1024 entries

Correct Answer: B)

Explanation: x86 IDT has exactly 256 entries (0-255), one per interrupt vector. Vectors 0-31 are reserved for CPU exceptions (divide error, page fault, etc.). Vectors 32-255 are available for hardware IRQs and software interrupts.

255. Which interrupt vector number represents the x86 page fault exception?

A) 0
B) 8
C) 14
D) 32

Correct Answer: C)

Explanation: x86 exception 14 = Page Fault. The CPU generates this when a page table entry is not present (valid bit=0), protection violation, or write to read-only page. Key x86 exceptions: 0=divide, 8=double fault, 13=GPF, 14=page fault.

256. Interrupt nesting means:

A) Two interrupts cancel each other
B) Higher-priority ISR can interrupt a lower-priority ISR
C) All interrupts wait in a queue
D) Interrupts merge together

Correct Answer: B)

Explanation: Interrupt nesting: while ISR for interrupt A runs, a higher-priority interrupt B arrives. If interrupts re-enabled in ISR A, B's ISR preempts A's ISR. After B's ISR completes, A's ISR resumes. Requires careful stack management.

■ HARD / TRICKY QUESTIONS

257. Assertion: The CPU automatically saves return address before executing ISR. Reason: Without saving PC, the interrupted program cannot resume after ISR completes. Select:

A) Both true; R explains A

B) Both true; R doesn't explain A

C) A true; R false

D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. CPU hardware automatically pushes PC (and flags/PSW) onto the stack before jumping to ISR. This allows IRET instruction at end of ISR to pop and restore PC, resuming the interrupted program exactly where it stopped.

---

258. On x86, the difference between INT n and SYSCALL instructions:

A) Both do exactly the same thing

B) INT n goes through IDT with full privilege check; SYSCALL is faster, bypasses IDT, uses MSR registers

C) SYSCALL is for hardware; INT for software

D) INT n is faster than SYSCALL

Correct Answer: B)

Explanation: INT n: goes through IDT, has overhead (descriptor check, privilege level transition, stack switching). SYSCALL (added in AMD64): uses MSR registers (STAR, LSTAR) for fast entry — no descriptor lookup. SYSCALL is ~30% faster for system call entry/exit on 64-bit systems.

DMA – ADVANCED & APPLICATION CONCEPTS

■ EASY LEVEL QUESTIONS

259. Which of the following devices commonly uses DMA?

A) CPU registers
B) Hard disk controller
C) Arithmetic Logic Unit
D) Program counter

Correct Answer: B)

Explanation: Hard disk (HDD/SSD) controllers use DMA to transfer large data blocks directly to/from RAM without CPU involvement. Also: network cards, sound cards, GPU, USB controllers, graphics cards.

---

260. DMA reduces CPU load by handling:

A) Arithmetic operations
B) Bulk data transfer between devices and memory
C) Cache management
D) Process scheduling

Correct Answer: B)

Explanation: DMA handles bulk data transfer (I/O ↔ memory), freeing CPU from byte-by-byte transfer loops. CPU can execute other computations concurrently, improving overall system throughput.

---

261. The DMA controller needs the system _____ to transfer data.

A) CPU registers
B) Clock signal only
C) Bus (address+data+control)
D) Interrupt line only

Correct Answer: C)

Explanation: DMA controller needs the system bus (address bus to specify memory location, data bus to transfer data, control bus for read/write signals) to move data between device and memory.

---

262. Bus mastering allows:

A) CPU to control all bus operations
B) I/O device to directly control the bus for DMA
C) Memory to initiate transfers
D) Cache to bypass bus

Correct Answer: B)

Explanation: Bus mastering: a PCI/PCIe device can become the bus master and directly initiate and control memory transfers without a central DMA chip. More efficient than legacy DMA channel approach.

■ MEDIUM LEVEL QUESTIONS

263. DMA is essential for which application?

A) Single-byte keyboard input
B) Large video frame capture
C) CPU arithmetic
D) Mouse movement

Correct Answer: B)

Explanation: Large video frame capture requires transferring megabytes per frame at 30-120fps — DMA is essential. Without DMA, CPU would be completely consumed by data movement, unable to process/encode frames in real-time.

---

264. The address used by DMA controller must be a _____ address.

A) Virtual
B) Logical
C) Physical
D) Segmented

Correct Answer: C)

Explanation: DMA controller uses physical (real) addresses — it bypasses the CPU's MMU. OS must translate virtual to physical addresses and pin (lock) those pages in RAM during DMA to prevent page swap-out.

---

265. Double buffering in DMA means:

A) Using two DMA controllers
B) Using two memory buffers — one being filled by DMA while other is processed by CPU
C) Doubling transfer speed
D) Using DMA twice

Correct Answer: B)

Explanation: Double buffering: DMA fills buffer A while CPU processes buffer B, then they swap. Enables true pipeline parallelism — no CPU wait for DMA and no DMA wait for CPU. Used in audio streaming, network I/O, video capture.

---

266. Which PC bus standard replaced ISA DMA with bus mastering?

A) ISA enhanced
B) PCI (Peripheral Component Interconnect)
C) RS-232
D) USB 1.0

Correct Answer: B)

Explanation: PCI bus (1992) introduced bus mastering DMA, replacing legacy ISA 8237 DMA chip. PCI devices can become bus masters for direct memory transfers. PCIe (2003) further extended this with serial point-to-point links.

■ HARD / TRICKY QUESTIONS

267. A DMA transfer of 64KB at 200MB/s takes:

A) 0.32 ms
B) 3.2 ms
C) 32 ms
D) 320 µs

Correct Answer: D)

Explanation: Time = Size / Rate = 64KB / 200MB/s = 65536 bytes / (200×10^6 bytes/s) = 65536/200,000,000 = 0.000328 s = 328 µs ≈ 320 µs. During this time, CPU is free to execute ~328,000 instructions at 1GHz.

---

268. Match DMA with application: P: Keyboard, Q: Network packet receive, R: Audio playback, S: GPU texture upload / 1: DMA critical for real-time, 2: DMA essential for throughput, 3: DMA not needed (1 byte), 4: DMA used for large GPU memory

A) P-3, Q-2, R-1, S-4

B) P-1, Q-3, R-2, S-4

C) P-4, Q-1, R-3, S-2

D) P-2, Q-4, R-3, S-1

Correct Answer: A)

Explanation: Keyboard(P-3): 1 byte/keypress — no DMA needed, uses interrupt. Network(Q-2): packets arrive at high rate — DMA essential for throughput. Audio(R-1): continuous samples at 44.1kHz — DMA critical for real-time. GPU texture(S-4): large data from RAM to GPU — DMA for PCIe bulk transfer.

MIXED REVISION – ALL TOPICS

■ EASY LEVEL QUESTIONS

269. Which of the following is NOT a property of a process?

A) Process ID
B) Memory space
C) Source code file
D) Process state

Correct Answer: C)

Explanation: Source code file is not a property of a running process. Process properties include: PID, process state, program counter, CPU registers, memory space, open files, I/O status. Source code is compiled before creating a process.

---

270. The ready queue in an OS contains processes that are:

A) Running on CPU
B) Waiting for I/O
C) Ready to run but waiting for CPU
D) Terminated

Correct Answer: C)

Explanation: Ready queue: processes that are loaded in memory, have all resources needed, and are waiting only for CPU allocation by the scheduler.

---

271. Page table is stored in:

A) CPU registers
B) Main memory (RAM)
C) Hard disk
D) Cache only

Correct Answer: B)

Explanation: Page tables are stored in main memory (RAM). The PTBR (Page Table Base Register) points to the page table in RAM. TLB caches recent page table entries for fast access.

---

272. Which scheduling algorithm uses time quantum?

A) FCFS
B) SJF
C) Round Robin
D) Priority (non-preemptive)

Correct Answer: C)

Explanation: Round Robin uses time quantum (time slice). Each process gets CPU for one quantum. FCFS, non-preemptive SJF, and non-preemptive priority do not use time quantum.

---

273. FAT file system was originally developed for:

A) Unix systems
B) MS-DOS on IBM PCs
C) Apple Macintosh
D) Network servers

Correct Answer: B)

Explanation: FAT was developed by Microsoft (originally by Marc McDonald and Bill Gates) for MS-DOS in 1977. FAT12 for floppy disks, FAT16 for early hard disks, FAT32 for larger disks introduced with Windows 95 OSR2.

---

274. A hardware interrupt is also called:

A) Trap
B) Exception
C) External interrupt
D) Software interrupt

Correct Answer: C)

Explanation: Hardware interrupts are also called external interrupts — they come from external hardware devices. Software interrupts are called traps or exceptions. The classification: External (hardware), Internal (exceptions), and Software (traps).

---

275. Virtual memory is transparent to:

A) The OS
B) The hardware
C) User programs (they see a large, contiguous address space)
D) Device drivers

Correct Answer: C)

Explanation: Virtual memory is transparent to user programs. Programs see a large, contiguous virtual address space. The OS and hardware (MMU) handle translation to physical addresses invisibly.

■ MEDIUM LEVEL QUESTIONS

276. Which of the following is an example of preemptive scheduling?

A) FCFS
B) Non-preemptive SJF
C) Round Robin
D) Non-preemptive Priority

Correct Answer: C)

Explanation: Round Robin is preemptive — processes are preempted after each time quantum. FCFS and non-preemptive SJF/Priority are non-preemptive — process runs to completion or I/O block.

---

277. A page size of 4KB means the offset field in a logical address is:

A) 10 bits
B) 12 bits
C) 16 bits
D) 4 bits

Correct Answer: B)

Explanation: Page size = 4KB = 4096 bytes = 2^12. Offset field = log2(4096) = 12 bits. The remaining bits in the logical address form the page number field.

---

278. The clock (second chance) page replacement algorithm uses:

A) Timestamps
B) Reference bit and circular pointer
C) LRU stack
D) Random selection

Correct Answer: B)

Explanation: Clock algorithm: circular list of pages with a reference bit. When replacement needed, pointer advances. Page with ref=1: clear bit, advance (second chance). Page with ref=0: replace. Cheap LRU approximation.

---

279. Memory-mapped files allow:

A) Files to be stored in DMA
B) Files to be accessed as if part of the process's address space
C) Files to be stored in CPU registers
D) Files to skip the file system

Correct Answer: B)

Explanation: Memory-mapped files (mmap() in Unix): file is mapped into process virtual address space. File I/O becomes memory access — no explicit read()/write() calls. OS handles paging between file and memory. Used for shared memory and large file processing.

---

280. Which of the following is true about DMA?

A) DMA requires CPU for every byte transferred
B) DMA controller accesses bus independently of CPU
C) DMA only works with software interrupts
D) DMA increases CPU load

Correct Answer: B)

Explanation: DMA controller can independently access the system bus to transfer data between device and memory. CPU initiates the transfer then is free. DMA completion is signaled to CPU via interrupt.

---

281. A system with 2GB RAM and 4GB virtual address space: the extra 2GB comes from:

A) CPU cache
B) Second CPU
C) Swap space on disk
D) ROM

Correct Answer: C)

Explanation: Extra virtual memory beyond physical RAM comes from swap space (paging file) on disk. Pages not in RAM are stored there and swapped in on demand (page fault). Disk is ~100,000x slower than RAM.

---

282. Which is the correct order of memory speed (fastest to slowest)?

A) Disk > RAM > Cache > Registers
B) Registers > Cache > RAM > Disk
C) Cache > Registers > RAM > Disk
D) RAM > Cache > Disk > Registers

Correct Answer: B)

Explanation: Speed order (fastest→slowest): CPU Registers (~1 cycle) > L1/L2/L3 Cache (~1-30 cycles) > RAM (~100 cycles / 100ns) > SSD (~100µs) > HDD (~10ms). Each level is ~10-100x slower than previous.

■ HARD / TRICKY QUESTIONS

283. A system has: 3 frames, reference string 1,2,3,1,2,4,1,2,3,4. Compare FIFO vs LRU page faults:

A) FIFO=6, LRU=6
B) FIFO=7, LRU=6
C) FIFO=6, LRU=5
D) FIFO=8, LRU=7

Correct Answer: B)

Explanation: FIFO: 1(F),2(F),3(F),1(H),2(H),4(F-evict1),1(F-evict2),2(F-evict3),3(F-evict4),4(F-evict1)=7 faults. LRU: 1(F),2(F),3(F),1(H),2(H),4(F-evict3),1(H),2(H),3(F-evict4),4(F-evict1)=6 faults. FIFO=7 > LRU=6.

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284. Process P1: AT=0,BT=3. P2: AT=2,BT=6. P3: AT=4,BT=4. P4: AT=6,BT=5. SRTF. Average TAT:

A) 9.25 ms
B) 8.75 ms
C) 10.50 ms
D) 7.00 ms

Correct Answer: B)

Explanation: SRTF: P1(0-2,rem1). P2 arrives(BT6>rem1→P1 continues). P1(2-3,done). At 3: P2(BT6). P2(3-4,rem5). P3 arrives(BT4<5→preempt). P3(4-6,rem2). P4 arrives(BT5>rem2→P3 continues). P3(6-8,done). At 8: P2(rem5),P4(BT5). Same BT, pick by arrival: P2(8-13,done). P4(13-18,done). TAT: P1=3, P2=13-2=11, P3=8-4=4, P4=18-6=12. Avg=(3+11+4+12)/4=30/4=7.5ms.

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285. Assertion: LRU approximates OPT better than FIFO. Reason: LRU uses recent past as predictor of near future, while FIFO ignores usage patterns entirely. Select:

A) Both true; R explains A
B) Both true; R doesn't explain A
C) A true; R false
D) A false; R true

Correct Answer: A)

Explanation: Both true and R explains A. LRU's use of recency (past access pattern) is a good predictor of future access (temporal locality). FIFO only considers load order, ignoring actual usage. This is why LRU consistently gives fewer page faults than FIFO.

MOCK TEST – FULL SIMULATION

Instructions: Attempt all 30 questions. Each correct answer = 1 mark. No negative marking. Time limit: 60 minutes.

MT1. The OS kernel runs in which mode?

A) User mode
B) Kernel/Supervisor mode
C) Guest mode
D) Debug mode

Correct Answer: B)

Explanation: Kernel runs in privileged mode, allowing unrestricted hardware access. User apps run in restricted user mode.

MT2. Convoy effect in FCFS is caused by:

A) Short processes
B) Long processes blocking shorter ones
C) Equal burst times
D) Different arrival times

Correct Answer: B)

Explanation: Convoy effect: a long process arriving before short ones forces all short processes to wait — like a slow convoy blocking faster vehicles.

MT3. Round Robin with Q=4ms, P1(BT=7ms) executes P1 how many times?

A) Once for 7ms
B) Twice (4ms + 3ms)
C) Three times
D) Once for 4ms only

Correct Answer: B)

Explanation: P1 gets CPU for 4ms (Q=4), remaining 3ms. Next turn: 3ms (< Q). Total: 2 turns.

MT4. Which page replacement has Belady's Anomaly?

A) OPT
B) LRU
C) FIFO
D) Clock

Correct Answer: C)

Explanation: FIFO can exhibit Belady's Anomaly (more frames → more faults). LRU and OPT are stack algorithms and immune.

MT5. FAT12 uses _____ bits for cluster numbers.

A) 8
B) 12
C) 16
D) 32

Correct Answer: B)

Explanation: FAT12 uses 12-bit cluster numbers, supporting max 4086 clusters. Used for small disks like floppy disks.

MT6. DMA cycle stealing transfers:

A) Entire block at once
B) One bus cycle at a time
C) Only when CPU idle
D) Using CPU for every byte

Correct Answer: B)

Explanation: Cycle stealing: DMA takes one bus cycle, transfers one data unit, releases bus to CPU, repeats.

MT7. Software interrupt (trap) switches CPU to:

A) User mode
B) Kernel mode
C) Idle mode
D) Debug mode

Correct Answer: B)

Explanation: Software trap/INT instruction switches CPU from user mode to kernel mode for system call execution.

MT8. Segmentation eliminates _____ fragmentation.

A) Internal
B) External
C) Both
D) Neither

Correct Answer: A)

Explanation: Segmentation eliminates internal fragmentation (segments exactly the size needed). It suffers external fragmentation.

MT9. Memory-Mapped I/O uses _____ instructions.

A) Special I/O (IN/OUT)
B) Regular memory (MOV/LOAD/STORE)
C) DMA
D) Interrupt

Correct Answer: B)

Explanation: MMIO uses normal memory instructions since device registers are in the memory address space.

MT10. Working set model prevents:

A) External fragmentation
B) Thrashing
C) Internal fragmentation
D) Deadlock

Correct Answer: B)

Explanation: Working set model prevents thrashing by ensuring each process's active pages are in memory before allocating CPU.

MT11. Priority scheduling: P1(P=5), P2(P=1), P3(P=3). Execution order (lower=higher priority):

A) P1,P2,P3
B) P2,P3,P1
C) P3,P1,P2
D) P1,P3,P2

Correct Answer: B)

Explanation: Lower priority number = higher priority. Order: P2(1) → P3(3) → P1(5).

MT12. TLB is used to speed up:

A) Disk access
B) Page table lookup
C) Context switching
D) Interrupt handling

Correct Answer: B)

Explanation: TLB is a fast associative cache for recent page→frame mappings, reducing effective memory access time.

MT13. FCFS scheduling is:

A) Preemptive
B) Non-preemptive
C) Priority-based
D) Time-quantum-based

Correct Answer: B)

Explanation: FCFS is non-preemptive — process runs to completion without preemption.

MT14. Hardware interrupts are generated by:

A) Software traps
B) External hardware devices
C) OS kernel
D) Compiler

Correct Answer: B)

Explanation: Hardware interrupts come from physical devices (keyboard, disk, timer, NIC) signaling the CPU.

MT15. PTBR (Page Table Base Register) contains:

A) Page size
B) Base address of current process's page table
C) Number of pages
D) TLB address

Correct Answer: B)

Explanation: PTBR stores the starting address of the current process's page table in memory. Updated on context switch.

MT16. If RR quantum > all burst times, it behaves like:

A) Priority scheduling
B) SJF
C) FCFS
D) Multilevel queue

Correct Answer: C)

Explanation: Quantum > all burst times → every process completes in first turn → no preemption → FCFS behavior.

MT17. DMA completion is signaled to CPU via:

A) Polling
B) System call
C) Interrupt
D) Memory write

Correct Answer: C)

Explanation: DMA controller sends interrupt to CPU on transfer completion. CPU's ISR processes the event.

MT18. Which scheduling is free from starvation?

A) Priority scheduling
B) Round Robin
C) SJF
D) Multilevel queue

Correct Answer: B)

Explanation: RR: every process gets CPU within (n-1)×quantum time — guaranteed, no starvation.

MT19. Virtual address 0x1A2B, page size 256 (0x100). Page# and offset:

A) Page=0x1A, Offset=0x2B
B) Page=0x1, Offset=0xA2B
C) Page=0x1A2, Offset=0xB
D) Page=0xA, Offset=0x2B

Correct Answer: A)

Explanation: Page size=256=2^8. Offset=lower 8 bits=0x2B. Page#=upper bits=0x1A.

MT20. Thrashing can be reduced by:

A) Increasing page size
B) Using working set model
C) Decreasing frames
D) Increasing time quantum

Correct Answer: B)

Explanation: Working set model prevents thrashing by tracking and maintaining each process's active working set in memory.

MT21. fseek(fp, 0, SEEK_END) moves file pointer to:

A) Beginning
B) Current position
C) End of file
D) Middle

Correct Answer: C)

Explanation: SEEK_END=from end of file. Offset=0 → exactly at end. Common use: find file size.

MT22. ISA DMA channel 4 is used as:

A) Audio
B) Floppy
C) Cascade
D) Memory refresh

Correct Answer: C)

Explanation: Channel 4 is the cascade connecting two 8237 DMA controllers, enabling 8 total channels.

MT23. Paging may cause _____ fragmentation.

A) External
B) Internal
C) Both
D) Neither

Correct Answer: B)

Explanation: Paging → internal fragmentation. Last page of process may not be fully used, wasting space inside the frame.

MT24. In segmentation, offset must be _____ the limit.

A) Equal to
B) Greater than
C) Less than
D) Greater than or equal to

Correct Answer: C)

Explanation: Offset < limit for valid access. Offset >= limit → segmentation fault (protection violation trap).

MT25. Which is NOT a valid CPU scheduling criterion?

A) CPU utilization
B) Throughput
C) Compilation speed
D) Response time

Correct Answer: C)

Explanation: Compilation speed is not a CPU scheduling criterion. Valid criteria: CPU utilization, throughput, turnaround time, waiting time, response time.

MT26. IRQ stands for:

A) Interrupt Request
B) Internal Register Queue
C) Input/Output Request
D) Interrupt Register

Correct Answer: A)

Explanation: IRQ = Interrupt Request. Hardware devices use IRQ lines to signal CPU. PIC manages and prioritizes these.

MT27. MMIO regions must be non-cacheable because:

A) It speeds up access
B) Device register values change asynchronously
C) CPU cannot cache them
D) It saves memory

Correct Answer: B)

Explanation: Hardware changes device registers independently. Caching causes stale values. Non-cacheable ensures fresh reads.

MT28. FCFS: P1(AT=0,BT=4), P2(AT=1,BT=2), P3(AT=2,BT=6). Average waiting time:

A) 2.33 ms
B) 3.33 ms
C) 4.00 ms
D) 1.67 ms

Correct Answer: A)

Explanation: Gantt: P1(0-4), P2(4-6), P3(6-12). WT: P1=0, P2=4-1=3, P3=6-2=4. Avg=(0+3+4)/3=7/3≈2.33ms.

MT29. LRU page replacement replaces the page:

A) Used most recently
B) Not used for the longest time (past)
C) Loaded longest ago (FIFO)
D) With lowest page number

Correct Answer: B)

Explanation: LRU: Least Recently Used. Replaces page not used for the longest time in the past — approximates OPT using history.

MT30. Which DMA mode causes NO CPU slowdown?

A) Burst mode
B) Cycle stealing
C) Transparent mode
D) Block mode

Correct Answer: C)

Explanation: Transparent DMA uses bus only when CPU doesn't need it (during internal ops). Zero CPU slowdown but slowest transfer.

MOCK TEST – ANSWER KEY

MT1: B	MT2: B	MT3: B	MT4: C	MT5: B	MT6: B
MT7: B	MT8: A	MT9: B	MT10: B	MT11: B	MT12: B
MT13: B	MT14: B	MT15: B	MT16: C	MT17: C	MT18: B
MT19: A	MT20: B	MT21: C	MT22: C	MT23: B	MT24: C
MT25: C	MT26: A	MT27: B	MT28: A	MT29: B	MT30: C

MASTER ANSWER KEY – ALL QUESTIONS

Q#	Section	Level	Answer
1	Operating System – Basics & Functions	Easy	**B**
2	Operating System – Basics & Functions	Easy	**C**
3	Operating System – Basics & Functions	Easy	**B**
4	Operating System – Basics & Functions	Easy	**C**
5	Operating System – Basics & Functions	Easy	**B**
6	Operating System – Basics & Functions	Easy	**C**
7	Operating System – Basics & Functions	Easy	**B**
8	Operating System – Basics & Functions	Easy	**C**
9	Operating System – Basics & Functions	Easy	**B**
10	Operating System – Basics & Functions	Easy	**B**
11	Operating System – Basics & Functions	Easy	**B**
12	Operating System – Basics & Functions	Easy	**B**
13	Operating System – Basics & Functions	Easy	**A**
14	Operating System – Basics & Functions	Medium	**C**
15	Operating System – Basics & Functions	Medium	**B**
16	Operating System – Basics & Functions	Medium	**C**
17	Operating System – Basics & Functions	Medium	**B**
18	Operating System – Basics & Functions	Medium	**C**
19	Operating System – Basics & Functions	Medium	**B**
20	Operating System – Basics & Functions	Medium	**B**
21	Operating System – Basics & Functions	Medium	**B**
22	Operating System – Basics & Functions	Medium	**C**
23	Operating System – Basics & Functions	Hard	**A**
24	Operating System – Basics & Functions	Hard	**B**
25	Operating System – Basics & Functions	Hard	**A**
26	Operating System – Basics & Functions	Hard	**B**
27	Process Management – FCFS Scheduling	Easy	**A**
28	Process Management – FCFS Scheduling	Easy	**B**
29	Process Management – FCFS Scheduling	Easy	**B**
30	Process Management – FCFS Scheduling	Easy	**B**
31	Process Management – FCFS Scheduling	Easy	**B**
32	Process Management – FCFS Scheduling	Easy	**B**
33	Process Management – FCFS Scheduling	Medium	**A**
34	Process Management – FCFS Scheduling	Medium	**C**
35	Process Management – FCFS Scheduling	Medium	**B**

36	Process Management – FCFS Scheduling	Medium	**B**
37	Process Management – FCFS Scheduling	Medium	**B**
38	Process Management – FCFS Scheduling	Hard	**A**
39	Process Management – FCFS Scheduling	Hard	**A**
40	Process Management – FCFS Scheduling	Hard	**B**
41	Process Management – Round Robin Scheduling	Easy	**B**
42	Process Management – Round Robin Scheduling	Easy	**B**
43	Process Management – Round Robin Scheduling	Easy	**C**
44	Process Management – Round Robin Scheduling	Easy	**B**
45	Process Management – Round Robin Scheduling	Easy	**B**
46	Process Management – Round Robin Scheduling	Easy	**B**
47	Process Management – Round Robin Scheduling	Easy	**C**
48	Process Management – Round Robin Scheduling	Medium	**A**
49	Process Management – Round Robin Scheduling	Medium	**B**
50	Process Management – Round Robin Scheduling	Medium	**B**
51	Process Management – Round Robin Scheduling	Medium	**C**
52	Process Management – Round Robin Scheduling	Medium	**C**
53	Process Management – Round Robin Scheduling	Hard	**A**
54	Process Management – Round Robin Scheduling	Hard	**A**
55	Process Management – Round Robin Scheduling	Hard	**A**
56	Process Management – Priority-Based Schedulin…	Easy	**C**
57	Process Management – Priority-Based Schedulin…	Easy	**B**
58	Process Management – Priority-Based Schedulin…	Easy	**B**
59	Process Management – Priority-Based Schedulin…	Easy	**C**
60	Process Management – Priority-Based Schedulin…	Easy	**C**
61	Process Management – Priority-Based Schedulin…	Easy	**A**
62	Process Management – Priority-Based Schedulin…	Medium	**C**
63	Process Management – Priority-Based Schedulin…	Medium	**B**
64	Process Management – Priority-Based Schedulin…	Medium	**A**
65	Process Management – Priority-Based Schedulin…	Medium	**B**
66	Process Management – Priority-Based Schedulin…	Medium	**B**
67	Process Management – Priority-Based Schedulin…	Hard	**A**
68	Process Management – Priority-Based Schedulin…	Hard	**A**
69	Process Management – Priority-Based Schedulin…	Hard	**A**
70	Memory Management – Paging	Easy	**B**
71	Memory Management – Paging	Easy	**B**
72	Memory Management – Paging	Easy	**C**
73	Memory Management – Paging	Easy	**B**
74	Memory Management – Paging	Easy	**B**

75	Memory Management – Paging	Easy	**B**
76	Memory Management – Paging	Medium	**B**
77	Memory Management – Paging	Medium	**B**
78	Memory Management – Paging	Medium	**C**
79	Memory Management – Paging	Medium	**A**
80	Memory Management – Paging	Medium	**B**
81	Memory Management – Paging	Medium	**A**
82	Memory Management – Paging	Hard	**A**
83	Memory Management – Paging	Hard	**A**
84	Memory Management – Paging	Hard	**A**
85	Memory Management – Segmentation	Easy	**B**
86	Memory Management – Segmentation	Easy	**B**
87	Memory Management – Segmentation	Easy	**C**
88	Memory Management – Segmentation	Easy	**B**
89	Memory Management – Segmentation	Easy	**B**
90	Memory Management – Segmentation	Medium	**B**
91	Memory Management – Segmentation	Medium	**C**
92	Memory Management – Segmentation	Medium	**B**
93	Memory Management – Segmentation	Hard	**A**
94	Memory Management – Segmentation	Hard	**A**
95	Memory Management – Virtual Memory	Easy	**B**
96	Memory Management – Virtual Memory	Easy	**C**
97	Memory Management – Virtual Memory	Easy	**B**
98	Memory Management – Virtual Memory	Easy	**C**
99	Memory Management – Virtual Memory	Easy	**B**
100	Memory Management – Virtual Memory	Easy	**B**
101	Memory Management – Virtual Memory	Medium	**A**
102	Memory Management – Virtual Memory	Medium	**A**
103	Memory Management – Virtual Memory	Medium	**B**
104	Memory Management – Virtual Memory	Medium	**B**
105	Memory Management – Virtual Memory	Medium	**A**
106	Memory Management – Virtual Memory	Hard	**B**
107	Memory Management – Virtual Memory	Hard	**A**
108	Memory Management – Virtual Memory	Hard	**A**
109	I/O Management – Concept of I/O Port	Easy	**B**
110	I/O Management – Concept of I/O Port	Easy	**B**
111	I/O Management – Concept of I/O Port	Easy	**B**
112	I/O Management – Concept of I/O Port	Easy	**B**
113	I/O Management – Concept of I/O Port	Easy	**B**

114	I/O Management – Concept of I/O Port	Easy	**B**
115	I/O Management – Concept of I/O Port	Medium	**B**
116	I/O Management – Concept of I/O Port	Medium	**B**
117	I/O Management – Concept of I/O Port	Medium	**B**
118	I/O Management – Concept of I/O Port	Medium	**B**
119	I/O Management – Concept of I/O Port	Medium	**B**
120	I/O Management – Concept of I/O Port	Hard	**A**
121	I/O Management – Concept of I/O Port	Hard	**A**
122	File Management – FAT & File Handling Functio…	Easy	**B**
123	File Management – FAT & File Handling Functio…	Easy	**B**
124	File Management – FAT & File Handling Functio…	Easy	**C**
125	File Management – FAT & File Handling Functio…	Easy	**A**
126	File Management – FAT & File Handling Functio…	Easy	**B**
127	File Management – FAT & File Handling Functio…	Easy	**B**
128	File Management – FAT & File Handling Functio…	Medium	**B**
129	File Management – FAT & File Handling Functio…	Medium	**C**
130	File Management – FAT & File Handling Functio…	Medium	**C**
131	File Management – FAT & File Handling Functio…	Medium	**B**
132	File Management – FAT & File Handling Functio…	Medium	**A**
133	File Management – FAT & File Handling Functio…	Medium	**B**
134	File Management – FAT & File Handling Functio…	Hard	**A**
135	File Management – FAT & File Handling Functio…	Hard	**A**
136	File Management – FAT & File Handling Functio…	Hard	**A**
137	Interrupts – Software & Hardware	Easy	**B**
138	Interrupts – Software & Hardware	Easy	**C**
139	Interrupts – Software & Hardware	Easy	**B**
140	Interrupts – Software & Hardware	Easy	**A**
141	Interrupts – Software & Hardware	Easy	**B**
142	Interrupts – Software & Hardware	Easy	**A**
143	Interrupts – Software & Hardware	Easy	**B**
144	Interrupts – Software & Hardware	Medium	**C**
145	Interrupts – Software & Hardware	Medium	**B**
146	Interrupts – Software & Hardware	Medium	**C**
147	Interrupts – Software & Hardware	Medium	**B**
148	Interrupts – Software & Hardware	Medium	**B**
149	Interrupts – Software & Hardware	Hard	**A**
150	Interrupts – Software & Hardware	Hard	**A**
151	Interrupts – Software & Hardware	Hard	**A**
152	I/O and Memory-Based Addressing	Easy	**B**

153	I/O and Memory-Based Addressing	Easy	**B**
154	I/O and Memory-Based Addressing	Easy	**B**
155	I/O and Memory-Based Addressing	Easy	**B**
156	I/O and Memory-Based Addressing	Easy	**B**
157	I/O and Memory-Based Addressing	Medium	**B**
158	I/O and Memory-Based Addressing	Medium	**B**
159	I/O and Memory-Based Addressing	Medium	**C**
160	I/O and Memory-Based Addressing	Medium	**B**
161	I/O and Memory-Based Addressing	Hard	**A**
162	I/O and Memory-Based Addressing	Hard	**A**
163	DMA – Direct Memory Access	Easy	**B**
164	DMA – Direct Memory Access	Easy	**B**
165	DMA – Direct Memory Access	Easy	**C**
166	DMA – Direct Memory Access	Easy	**B**
167	DMA – Direct Memory Access	Easy	**B**
168	DMA – Direct Memory Access	Easy	**B**
169	DMA – Direct Memory Access	Medium	**B**
170	DMA – Direct Memory Access	Medium	**B**
171	DMA – Direct Memory Access	Medium	**C**
172	DMA – Direct Memory Access	Medium	**B**
173	DMA – Direct Memory Access	Medium	**B**
174	DMA – Direct Memory Access	Medium	**B**
175	DMA – Direct Memory Access	Hard	**A**
176	DMA – Direct Memory Access	Hard	**A**
177	DMA – Direct Memory Access	Hard	**A**
178	DMA – Direct Memory Access	Hard	**A**
179	Numerical & Calculation MCQs	Medium	**B**
180	Numerical & Calculation MCQs	Medium	**A**
181	Numerical & Calculation MCQs	Medium	**B**
182	Numerical & Calculation MCQs	Medium	**C**
183	Numerical & Calculation MCQs	Medium	**A**
184	Numerical & Calculation MCQs	Medium	**B**
185	Numerical & Calculation MCQs	Hard	**A**
186	Numerical & Calculation MCQs	Hard	**A**
187	Assertion-Reason & Tricky MCQs	Medium	**A**
188	Assertion-Reason & Tricky MCQs	Medium	**B**
189	Assertion-Reason & Tricky MCQs	Medium	**B**
190	Assertion-Reason & Tricky MCQs	Medium	**B**
191	Assertion-Reason & Tricky MCQs	Hard	**B**

192	Assertion-Reason & Tricky MCQs	Hard	**A**
193	Assertion-Reason & Tricky MCQs	Hard	**A**
194	Fill-in-the-Blank & One-Liner MCQs	Easy	**C**
195	Fill-in-the-Blank & One-Liner MCQs	Easy	**B**
196	Fill-in-the-Blank & One-Liner MCQs	Easy	**C**
197	Fill-in-the-Blank & One-Liner MCQs	Easy	**B**
198	Fill-in-the-Blank & One-Liner MCQs	Easy	**B**
199	Fill-in-the-Blank & One-Liner MCQs	Easy	**B**
200	Fill-in-the-Blank & One-Liner MCQs	Medium	**B**
201	Fill-in-the-Blank & One-Liner MCQs	Medium	**B**
202	Fill-in-the-Blank & One-Liner MCQs	Medium	**B**
203	Fill-in-the-Blank & One-Liner MCQs	Medium	**C**
204	Fill-in-the-Blank & One-Liner MCQs	Medium	**C**
205	Fill-in-the-Blank & One-Liner MCQs	Medium	**B**
206	OS Basics – System Calls & Kernel Types	Easy	**B**
207	OS Basics – System Calls & Kernel Types	Easy	**B**
208	OS Basics – System Calls & Kernel Types	Easy	**B**
209	OS Basics – System Calls & Kernel Types	Easy	**A**
210	OS Basics – System Calls & Kernel Types	Easy	**B**
211	OS Basics – System Calls & Kernel Types	Easy	**B**
212	OS Basics – System Calls & Kernel Types	Medium	**C**
213	OS Basics – System Calls & Kernel Types	Medium	**B**
214	OS Basics – System Calls & Kernel Types	Medium	**B**
215	OS Basics – System Calls & Kernel Types	Medium	**B**
216	OS Basics – System Calls & Kernel Types	Hard	**A**
217	OS Basics – System Calls & Kernel Types	Hard	**A**
218	Process Management – SJF & SRTF	Easy	**A**
219	Process Management – SJF & SRTF	Easy	**C**
220	Process Management – SJF & SRTF	Easy	**B**
221	Process Management – SJF & SRTF	Easy	**B**
222	Process Management – SJF & SRTF	Medium	**A**
223	Process Management – SJF & SRTF	Medium	**A**
224	Process Management – SJF & SRTF	Medium	**B**
225	Process Management – SJF & SRTF	Hard	**A**
226	Memory Management – Additional Concepts	Easy	**C**
227	Memory Management – Additional Concepts	Easy	**B**
228	Memory Management – Additional Concepts	Easy	**B**
229	Memory Management – Additional Concepts	Easy	**B**
230	Memory Management – Additional Concepts	Easy	**B**

231	Memory Management – Additional Concepts	Easy	C
232	Memory Management – Additional Concepts	Medium	C
233	Memory Management – Additional Concepts	Medium	B
234	Memory Management – Additional Concepts	Medium	C
235	Memory Management – Additional Concepts	Medium	B
236	Memory Management – Additional Concepts	Hard	A
237	Memory Management – Additional Concepts	Hard	A
238	File Management – Extended Concepts	Easy	B
239	File Management – Extended Concepts	Easy	B
240	File Management – Extended Concepts	Easy	B
241	File Management – Extended Concepts	Easy	B
242	File Management – Extended Concepts	Easy	B
243	File Management – Extended Concepts	Medium	D
244	File Management – Extended Concepts	Medium	B
245	File Management – Extended Concepts	Medium	A
246	File Management – Extended Concepts	Medium	B
247	File Management – Extended Concepts	Hard	C
248	File Management – Extended Concepts	Hard	A
249	Interrupts – Advanced Concepts	Easy	B
250	Interrupts – Advanced Concepts	Easy	C
251	Interrupts – Advanced Concepts	Easy	B
252	Interrupts – Advanced Concepts	Easy	B
253	Interrupts – Advanced Concepts	Easy	B
254	Interrupts – Advanced Concepts	Medium	B
255	Interrupts – Advanced Concepts	Medium	C
256	Interrupts – Advanced Concepts	Medium	B
257	Interrupts – Advanced Concepts	Hard	A
258	Interrupts – Advanced Concepts	Hard	B
259	DMA – Advanced & Application Concepts	Easy	B
260	DMA – Advanced & Application Concepts	Easy	B
261	DMA – Advanced & Application Concepts	Easy	C
262	DMA – Advanced & Application Concepts	Easy	B
263	DMA – Advanced & Application Concepts	Medium	B
264	DMA – Advanced & Application Concepts	Medium	C
265	DMA – Advanced & Application Concepts	Medium	B
266	DMA – Advanced & Application Concepts	Medium	B
267	DMA – Advanced & Application Concepts	Hard	D
268	DMA – Advanced & Application Concepts	Hard	A
269	Mixed Revision – All Topics	Easy	C

270	Mixed Revision – All Topics	Easy	**C**
271	Mixed Revision – All Topics	Easy	**B**
272	Mixed Revision – All Topics	Easy	**C**
273	Mixed Revision – All Topics	Easy	**B**
274	Mixed Revision – All Topics	Easy	**C**
275	Mixed Revision – All Topics	Easy	**C**
276	Mixed Revision – All Topics	Medium	**C**
277	Mixed Revision – All Topics	Medium	**B**
278	Mixed Revision – All Topics	Medium	**B**
279	Mixed Revision – All Topics	Medium	**B**
280	Mixed Revision – All Topics	Medium	**B**
281	Mixed Revision – All Topics	Medium	**C**
282	Mixed Revision – All Topics	Medium	**B**
283	Mixed Revision – All Topics	Hard	**B**
284	Mixed Revision – All Topics	Hard	**B**
285	Mixed Revision – All Topics	Hard	**A**